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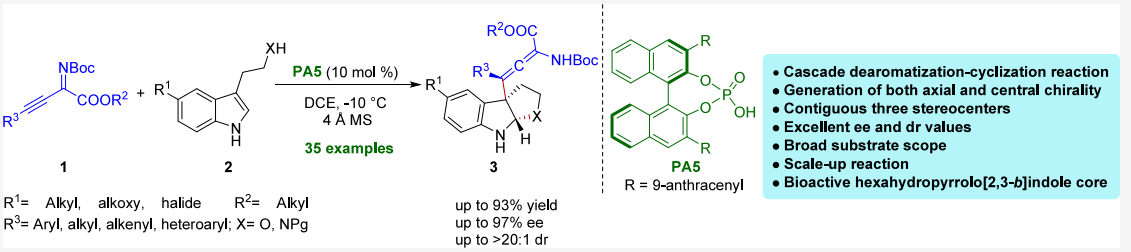
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Asymmetric Cascade Dearomatization−Cyclization Reaction of Tryptamines with β,γ-Alkynyl-α-imino Esters: Access to Hexahydropyrrolo[2,3-*b*]indole-Containing Tetrasubstituted α-Amino Allenoates

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ABSTRACT: An organocatalytic enantio- and diastereoselective synthesis of hexahydropyrrolo[2,3-*b*]indole-containing tetrasubstituted α-amino allenoates, exhibiting both axial and central chirality, has been accomplished via cascade dearomatization−cyclization reaction. The γ-addition to β,γ-alkynyl-α-imino esters provides a library of densely substituted highly enantioenriched allenes in high yields and excellent stereoselectivities. In addition, the scope of this methodology has been extended to tryptophol as well. A scale-up reaction and synthetic transformations of the products were performed to demonstrate the practical usefulness of this approach.

INTRODUCTION

The synthesis of enantioenriched heterocycles originating from indole stands as a pivotal objective in both organic and medicinal chemistry.[1] The hexahydropyrrolo[2,3-*b*]indole (HPI) core represents a class of indole alkaloids, featuring an all-carbon stereogenic center at the pseudobenzylic position. The hexahydropyrrolo[2,3-*b*]indole ring having a carbon substituent at the C-3a position is the defining structural feature for a range of bioactive natural products and pharmaceuticals including antifungal agent (**I**), antibacterial activity (**II**), cholinesterase inhibitor (**III**), and neurokinin antagonist (**IV**) (Figure 1).[2] Consequently, owing to the structural complexity and promising medicinal value, the synthesis of these heterocycles has drawn considerable attention among synthetic chemists over the past decade.[3] In this context, the frequently employed method for synthesizing chiral HPI frameworks relies on catalytic asymmetric dearomatization (CADA) of tryptamine derivatives, which can utilize the C-3 nucleophilicity of the indole ring to attack electrophiles (E) and initiate a cascade cyclization.[4]

In the past decade, there has been a notable emphasis on employing organoCADA of tryptamines with a diverse range of electrophiles.[5−8] However, alkyne derivatives have rarely been explored as electrophiles in the CADA reactions with tryptamines. The latter class of reaction helps synthesize HPIs containing an allene functionality at the C-3a position, thereby introducing both axial and central chirality. Axially chiral allenes are widespread and featured in many biological natural products, functional materials, and important synthetic intermediates.[9] Accordingly, considerable attention has been directed toward exploring effective methodologies for synthesizing densely functionalized allenic compounds.[10] Recently, Shi et al.[11a] and Li et al.[11b] utilized *para*-aminophenyl propargylic alcohols and α-(7-indolyl)propargylic alcohols, respectively, as alkyne derivatives with tryptamines for the asymmetric construction of axially chiral-tetrasubstituted allenes bearing a HPI unit (Scheme 1a). However, poor to moderate stereoselectivities were observed in most cases. Thus, despite several efforts made, efficient asymmetric synthesis of hexahydropyrrolo[2,3-*b*]indole bearing axially chiral tetrasubstituted allenes is still highly desirable.

Amidst various alkyne derivatives, β,γ-alkynyl-α-imino esters are one of the most potentially useful and challenging

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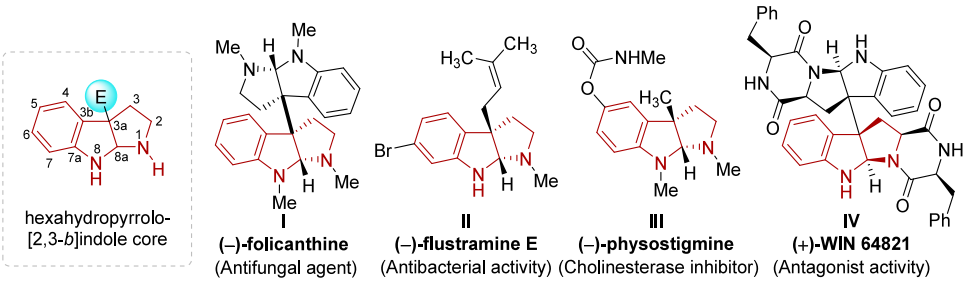
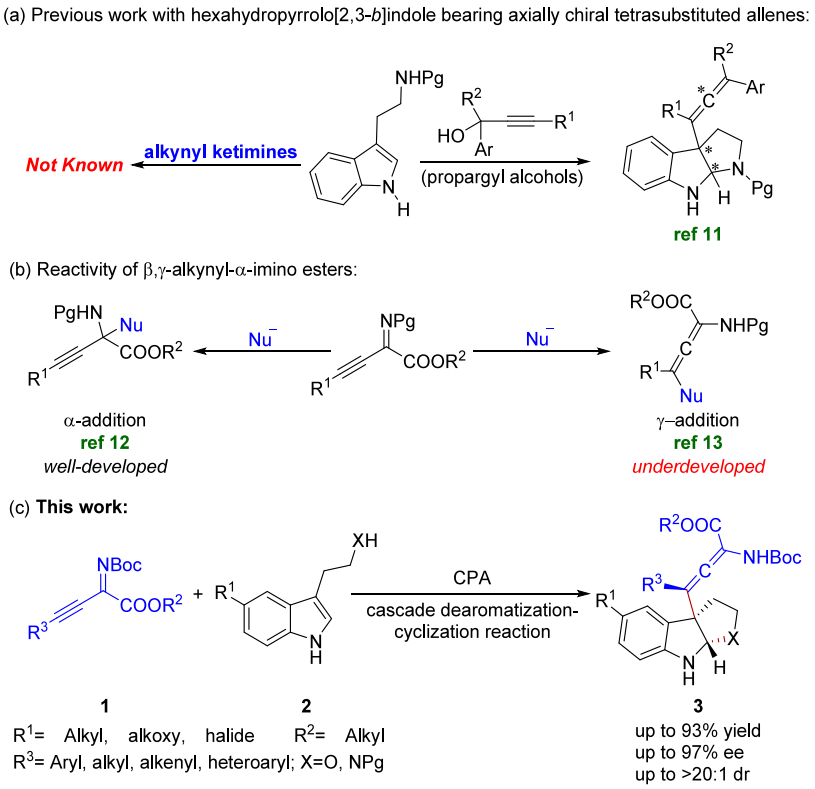


Figure 1. Selected C-3-alkylated hexahydropyrrolo[2,3-*b*]indole alkaloids.

Scheme 1. Reactions of Tryptamines and *β,γ*-Alkynyl-*α*-imino Esters



electrophiles due to their multiple reactive sites (*α*-addition vs *γ*-addition) (Scheme 1b). Notably, direct *α*-addition to *β,γ*-alkynyl-*α*-imino esters has been well explored for the synthesis of optically active alkynyl *α*-amino acid derivatives.[12] In sharp contrast, *γ*-addition of nucleophiles to *β,γ*-alkynyl-*α*-imino esters has been rare. To date, 2,3-disubstituted indoles,[13a] 1-substituted *β*-naphthols,[13b] isoxazolones,[13c] and pyrrolo[2,1-*a*]isoquinolines[13d] were employed successfully for synthesizing optically active amino allenoates. However, to the best of our knowledge, the cascade dearomatization−cyclization reaction with alkynyl ketimines as electrophiles has not been reported in the literature so far and demands attention. This is probably due to its formidable synthetic challenge, which includes (a) controlling the regioselectivity in *γ*-addition rather than *α*-addition and (b) controlling the diastero- and enantioselectivity in constructing HPIs bearing three contiguous stereogenic centers.

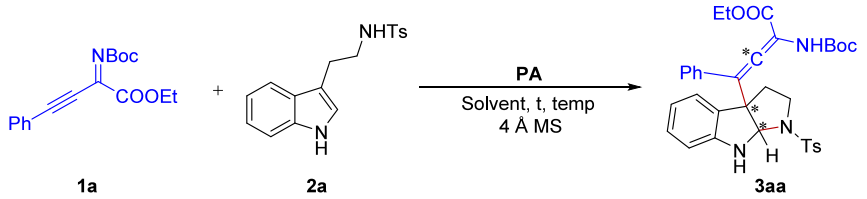
In continuation of our work on Brønsted acid-catalyzed enantioselective synthesis,[14] we became interested in exploring organocatalytic asymmetric dearomatization (CADA)-initiated ring closure reaction of tryptamines with *β,γ*-alkynyl-*α*-imino esters leading to the formation of tetrasubstituted *α*-amino allenoates containing an HPI unit (Scheme 1c). The incorporation of two potent pharmacophore units, namely, tetrasubstituted allene and pyrroloindoline, in a single chiral molecule would be interesting in the context of drug discovery.

RESULTS AND DISCUSSION

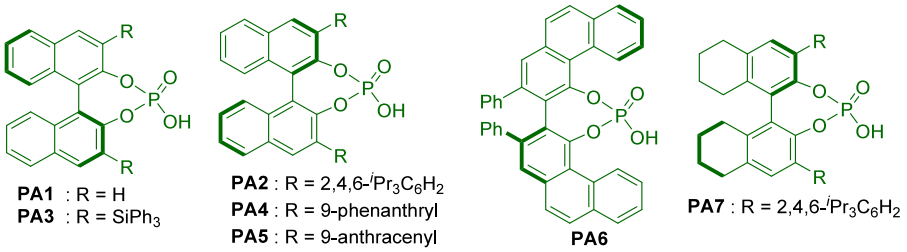
To check the feasibility of our hypothesis, we initiated our investigation by reacting *β,γ*-alkynyl-*α*-imino ester (**1a**) and *N*-tosyl-protected tryptamine (**2a**), in the presence of 10 mol % (*R*)-BINOL-derived phosphoric acid **PA1** with 4 Å molecular sieves (MS) in 1,2-dichloroethane at ambient temperature. The corresponding adduct **3aa** was isolated in 99% yield albeit with 2:1 dr and 12% ee for the major diastereomer (Table 1, entry 1). This preliminary result encouraged us to examine other chiral phosphoric acid catalysts with different backbones and steric environments (Table 1, entries 2−7). The optimal result was obtained using anthracenyl-substituted BINOL-derived catalyst **PA5**, as **3aa** was isolated in 98% yield with >20:1 dr and 91% ee (Table 1, entry 5). To further improve the enantioselectivity, we turned our attention to optimize

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Table 1. Optimization of Reaction Conditions[a]



1a 2a 3aa



PA1 : R = H
PA3 : R = $SiPh_3$

PA2 : R = 2,4,6-$^{i}Pr_3C_6H_2$
PA4 : R = 9-phenanthryl
PA5 : R = 9-anthracenyl

PA6

PA7 : R = 2,4,6-$^{i}Pr_3C_6H_2$

entry	catalyst	solvent	t (h)	yield (%)[b]	ee (%)[c]	dr[d]
1	PA1	DCE	5	99	−12	2:1
2	PA2	DCE	36	35	54	>20:1
3	PA3	DCE	36	38	−53	7:1
4	PA4	DCE	12	98	75	>20:1
5	PA5	DCE	8	98	91	>20:1
6	PA6	DCE	12	95	12	5:1
7	PA7	DCE	36	35	75	>20:1
8	PA5	CH_2Cl_2	8	89	91	>20:1
9	PA5	$CHCl_3$	8	90	88	>20:1
10	PA5	CCl_4	24	90	83	>20:1
11	PA5	toluene	36	85	70	11:1
12	PA5	benzene	24	83	70	10:1
13	PA5	*p*-xylene	36	84	63	8:1
14	PA5	THF	48	NR		
15[e]	PA5	DCE	36	66	57	>20:1
16[f]	PA5	DCE	36	53	91	>20:1
17[g]	PA5	DCE	8	98	92	19:1
18[h]	PA5	DCE	8	92	94	>20:1
19[i]	PA5	DCE	10	92	96	>20:1
20[j]	PA5	DCE	16	81	96	>20:1

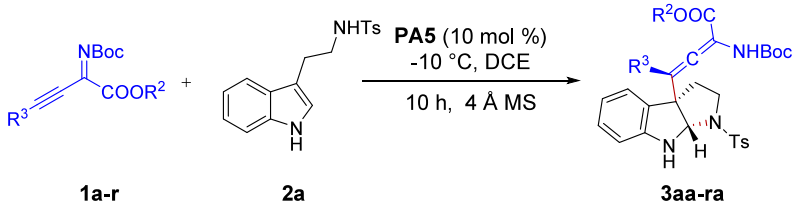
[a]Reaction conditions: room temperature of the laboratory during reactions was about 20−25 °C, **1a** (0.1 mmol), **2a** (0.12 mmol), chiral phosphoric acid (10 mol %), solvent (2 mL), RT with 4 Å MS (100 mg), unless noted otherwise. [b]Isolated yield. [c]Enantiomeric excess (ee) was determined by chiral HPLC analysis. [d]dr was determined from ^{1}H NMR of the crude reaction mixture. [e]Without 4 Å MS. [f]5 mol % catalyst loading. [g]20 mol % catalyst loading. [h]Reaction performed at 0 °C. [i]Reaction performed at −10 °C. [j]Reaction performed at −20 °C. PA = phosphoric acid, NR = no reaction, DCE = 1,2-dichloroethane, THF = tetrahydrofuran.

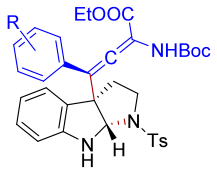
other reaction parameters. Initially, various polar and nonpolar solvents were evaluated (Table 1, entries 8−14). No enhancement in the enantioselectivity was observed, implying that DCE is the optimal solvent. Noticeably, carrying out the reaction in the absence of 4 Å MS led to an erosion of enantioselectivity (66% yield, 57% ee, and >20:1 dr; Table 1, entry 15). When the catalyst loading was reduced to 5 mol %, the stereoselectivity remained the same but yield decreased significantly (53% yield, 91% ee, and >20:1 dr; Table 1, entry 16). Upon increasing the catalyst loading to 20 mol %, enantioselectivity slightly improved; however, diastereoselectivity decreased (98% yield, 92% ee, and 19:1 dr, Table 1, entry 17). In our continued efforts toward getting improved results, the reaction was screened at various temperatures ranging from 25 to −20 °C (Table 1, entries 18−20). The optimal result was obtained at −10 °C (92% yield, 96% ee, and >20:1 dr; Table 1, entry 19).

With the reaction conditions established, a range of β,γ-alkynyl-α-imino esters **1** and tryptamines **2** were investigated to evaluate the substrate generality of the proposed CADA−cyclization reaction. Initially, several β,γ-alkynyl-α-imino esters (**1a−r**) were subjected to our optimized reaction condition, affording a variety of allene-substituted HPIs (**3aa−ra**) in high yields and excellent stereoselectivities (Scheme 2). In the beginning, alkynyl imino esters having a *para*-substitution on the aryl ring were tested and the corresponding products (**3aa−ja**) were obtained in moderate to good yields (up to 92%) with excellent diastereoselectivities (>20:1) and good to excellent enantioselectivities (up to 96%). However, lower yield and enantioselectivity were observed in the case of **3da** due to the presence of a bulky ^{t}Bu group at the *para* position (67% yield, 87% ee, and >20:1 dr). An increase in the chain length, in substrate **1g**, afforded product **3ga**, with slightly decreased yield but excellent stereoselectivity (74% yield, 93% ee, and >20:1 dr). Further, *o*-Me and *m*-Me substituents on the

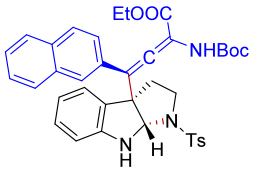
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Scheme 2. Substrate Scope of Alkynyl Imino Esters[a–d]

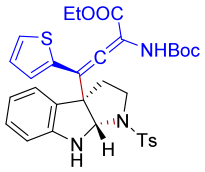




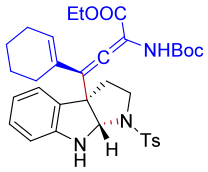
3aa: R= H; 92% yield, 96% ee, >20:1 dr
3ba: R= 4-OMe; 85% yield, 90% ee, >20:1 dr
3ca: R= 4-Me; 88% yield, 90% ee, >20:1 dr
3da: R= 4-^{t}Bu; 67% yield, 87% ee, >20:1 dr
3ea: R= 4-Ph; 88% yield, 90% ee, >20:1 dr
3fa: R= 4-Et; 85% yield, 92% ee, >20:1 dr
3ga: R= 4-pentyl; 74% yield, 93% ee, >20:1 dr
3ha: R= 4-F; 91% yield, 91% ee, >20:1 dr
3ia: R= 4-Cl; 90% yield, 92% ee, >20:1 dr
3ja: R= 4-Br; 83% yield, 90% ee, >20:1 dr
3ka: R= 2-Me; 87% yield, 93% ee, >20:1 dr
3la: R= 3-Me; 91% yield, 94% ee, >20:1 dr
3ma: R= 3,5-Me_2; 82% yield, 85% ee, >20:1 dr



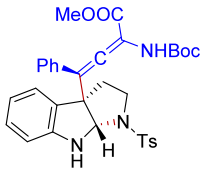
3na: 85% yield, 93% ee, >20:1 dr



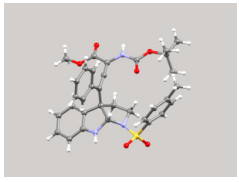
3oa: 86% yield, 90% ee, >20:1 dr



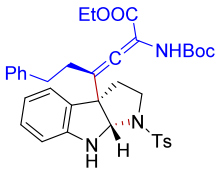
3pa: 81% yield, 83% ee, >20:1 dr



3qa: 92% yield, 94% ee, >20:1 dr



3qa: **CCDC** 2350171



3ra: 56% yield, 95% ee, >20:1 dr

[a]Reaction conditions: **1a** (0.1 mmol), **2a** (0.12 mmol), chiral phosphoric acid (10 mol %), solvent (2 mL), −10 °C with 4 Å MS (100 mg). [b]Isolated yield of the major diastereomer. [c]Enantiomeric excess (ee) was determined by chiral HPLC analysis. [d]dr was determined from ^{1}H NMR of the crude reaction mixture.

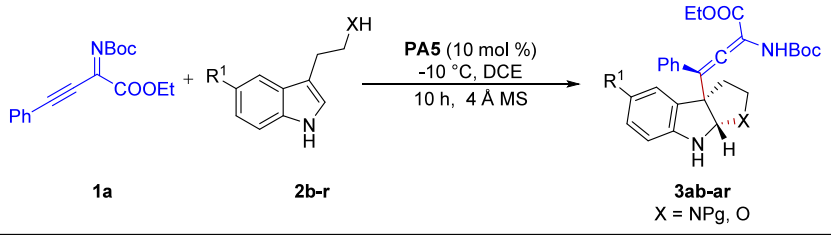
phenyl ring of the alkynyl imino ester (**1k**,**1l**) also worked well to furnish products **3ka** (87% yield, 93% ee, and >20:1 dr) and **3la** (91% yield, 94% ee, and >20:1 dr), respectively. Surprisingly, disubstitution on the phenyl ring in **1m** resulted in product **3ma** with moderate enantioselectivity (82% yield, 85% ee, and >20:1 dr). The reaction proceeded smoothly when the phenyl ring was replaced by a bulky naphthyl (**1n**) or heteroaryl thienyl (**1o**) group, yielding **3na** (85% yield, 93% ee, and >20:1 dr) and **3oa** (86% yield, 90% ee, and >20:1 dr), respectively, in moderate yields with excellent stereoselectivities. Notably, the imino ester containing an alkenyl moiety **1p** readily underwent the reaction, to produce **3pa** in good yield, excellent diastereoselectivity, and moderate enantioselectivity (81% yield, 83% ee, and >20:1 dr). In addition, the reaction condition was also applicable with alkynyl ketimino ester bearing COOMe substitution, producing the corresponding product **3qa** (92% yield, 94% ee, and >20:1 dr). In the case of an alkyl substituent in **1r**, product **3ra** was obtained with excellent stereoselectivity yet lower yield (56% yield, 95% ee, and >20:1 dr).

The scope of the CADA−cyclization reaction was further extended to various substituted tryptamines (**2a**−**q**), which afforded a range of structurally diverse allene-substituted pyrroloindolines (**3ab**−**aq**) in good to moderate yields (up to 93%) and excellent stereoselectivities (up to 97% ee and >20:1 dr) (Scheme 3). Representative *N*-tosyl-protected tryptamines with either electron-withdrawing or electron-donating groups at the 5-position resulted in smooth conversion and yielded HPI derivatives (**3ab**−**af**) in moderate yields (62−84%) with excellent stereoselectivities (87−95% ee and >20:1 dr). However, in the case of **3af**, with a Br group at the C-5 position, moderate yield and enantioselectivity were observed (62% yield, 87% ee, and >20:1 dr). Tryptamines containing different protecting groups were also tested in this protocol. Variation of electronically and sterically diverse sulfonyl-protecting groups (**2g**−**p**) revealed that the reaction condition was well tolerated in most of the cases and the desired products (**3ag**−**ap**) were obtained in low to good yields (45−93%) and excellent stereoselectivities (90−97% ee and >20:1 dr). It was discovered that better enantioselectivity is realized with aryl sulfonyls containing electron-donating groups rather than electron-withdrawing groups. The best outcome was seen in **3ao**, having 2,4,6-*iso*-propyl substitution at the phenyl ring (81% yield, 97% ee, and >20:1 dr). Nevertheless, in the case of **3am**, the introduction of a highly deactivating NO_2 group subsided the yield to 45%, although the stereoselectivity remained excellent (91% ee and >20:1 dr). The reaction protocol was also applicable with tryptamine bearing Boc protection, producing the corresponding product **3aq** (73% yield, 94% ee, and >20:1 dr). In our attempt to extend the scope of the methodology, tryptophol was also examined under our optimized reaction conditions. The corresponding product **3ar** with a furoindoline core bearing tetrasubstituted allene was obtained in slightly lower yield, moderate enantioselectivity, and excellent diastereoselectivity (66% yield, 87% ee, and >20:1 dr).

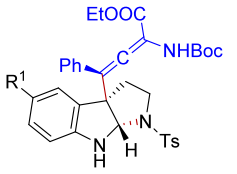
Next, a series of control experiments were performed to gain more insight into the reaction pathway (Scheme 4). At first, alkynyl trifluoromethyl ketimine **4** was allowed to react with *N*-tosyl-protected tryptamine **2a** under the optimized reaction

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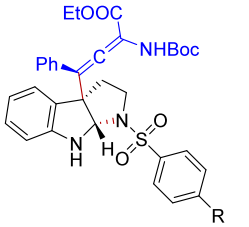
Scheme 3. Substrate Scope of Tryptamines[a–d]



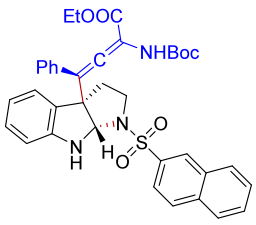
1a 2b-r 3ab-ar
X = NPg, O



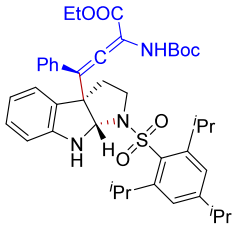
3ab: R^1 = OMe; 78% yield, 95% ee, >20:1 dr
3ac: R^1 = Me; 75% yield, 95% ee, >20:1 dr
3ad: R^1 = F; 84% yield, 95% ee, >20:1 dr
3ae: R^1 = Cl; 82% yield, 94% ee, >20:1 dr
3af: R^1 = Br; 62% yield, 87% ee, >20:1 dr



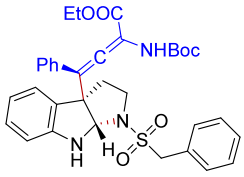
3ag: R = H; 85% yield, 94% ee, >20:1 dr
3ah: R = OMe; 90% yield, 94% ee, >20:1 dr
3ai: R = ^{t}Bu; 87% yield, 95% ee, >20:1 dr
3aj: R = Ph; 92% yield, 94% ee, >20:1 dr
3ak: R = F; 88% yield, 93% ee, >20:1 dr
3al: R = Br; 85% yield, 90% ee, >20:1 dr
3am: R = NO_2; 45% yield, 91% ee, >20:1 dr



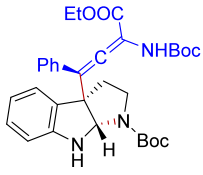
3an: 92% yield, 95% ee, >20:1 dr



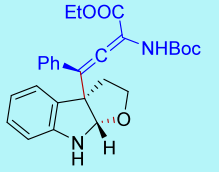
3ao: 81% yield, 97% ee, >20:1 dr



3ap: 93% yield, 95% ee, >20:1 dr



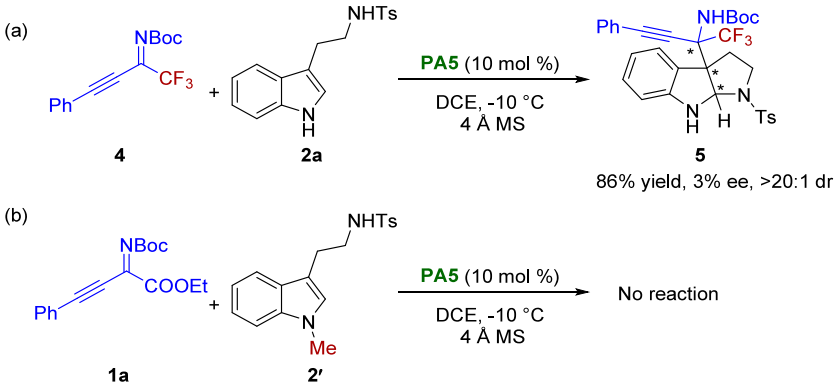
3aq: 73% yield, 94% ee, >20:1 dr



3ar: 66% yield, 87% ee, >20:1 dr
furoindoline core

[a]Reaction conditions: **1a** (0.1 mmol), **2a** (0.12 mmol), chiral phosphoric acid (10 mol %), solvent (2 mL), −10 °C with 4 Å MS (100 mg). [b]Isolated yield of the major diastereomer. [c]Enantiomeric excess (ee) was determined by chiral HPLC analysis. [d]dr was determined from ^{1}H NMR of the crude reaction mixture.

Scheme 4. Control Experiments



conditions (Scheme 4a). This resulted in the formation of the α-addition product **5** with nil enantioselectivity, suggesting that the COOR group of the β,γ-alkynyl-α-imino ester plays a crucial role in determining the regio- and enantioselectivity of the reaction. This can be attributed to the hydrogen bonding interaction of the carbonyl group with the CPA, which possibly generates spatial steric hindrance in the transition state. Furthermore, β,γ-alkynyl-α-imino ester **1a** was employed to react with N_a-protected tryptamine **2′**, but the reaction failed to give the product. The outcome shows that the free amino group of tryptamine is crucial for the cascade reaction to occur, possibly by forming an H-bond with the catalyst (Scheme 4b).

The absolute configuration of **3qa** (CCDC 2350171) was determined as (R_a, *R*, *S*) by X-ray crystallographic studies, and the stereochemistry of other chiral allene-substituted HPIs was assigned by analogy. Based on the crystallographic analysis, control experiments, and previous reports,[11b,13a] a possible transition state is proposed (Figure 2). At first, the bifunctional phosphoric acid catalyst **PA5** activates both β,γ-alkynyl-α-imino ester **1a** and tryptamine **2a** simultaneously via hydrogen

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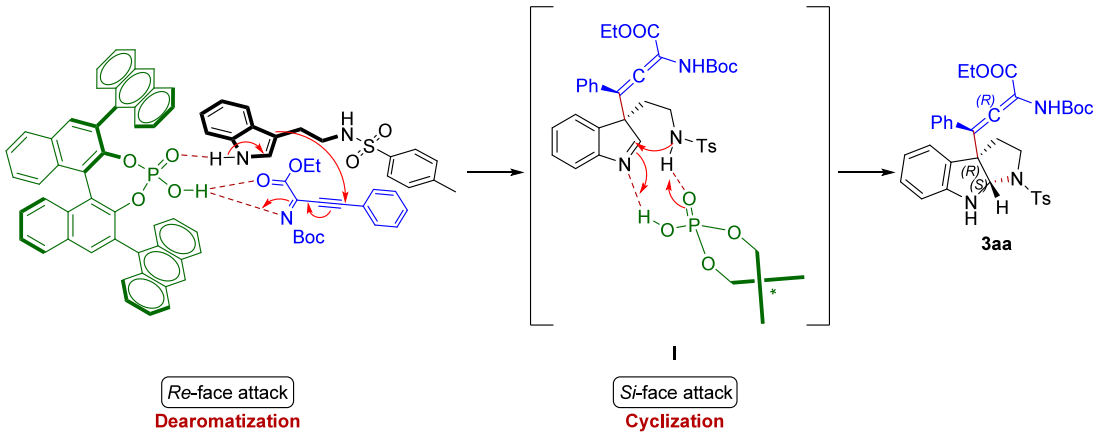
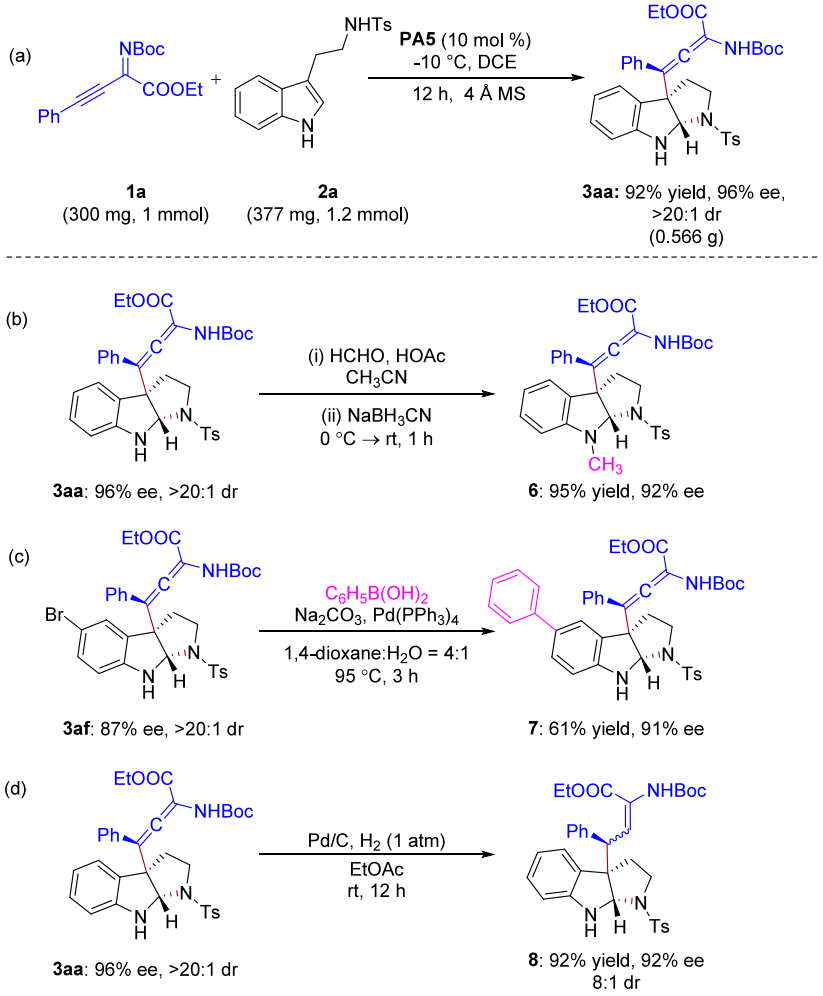


Figure 2. Proposed transition state model.

Scheme 5. Scale-up Reaction and Synthetic Transformations



bonding interaction. This facilitates dearomatization of tryptamine **2a**, and γ-addition takes place at the alkynyl imino ester **1a** from the *Re*-face of the nucleophile. The steric hindrance of the substituent at the C3 position of the indole plays a vital role in determining the regioselectivity (α/γ-addition) of the reaction. Furthermore, the chiral backbone of catalyst **PA5** controls the stereoselectivity through steric effects and $\pi-\pi$ interactions. Subsequently, a catalyst-assisted stereoselective intramolecular cyclization occurs to the *Si*-face of the 3,3-disubstituted indolenine intermediate **I**, to deliver a hexahydropyrrolo[2,3-*b*]indole core containing tetrasubstituted allene **3aa**.

Later, a scale-up reaction was carried out to showcase the practical utility of this methodology. Product **3aa** was isolated without any loss in yield or stereoselectivities (92% yield, 96% ee, >20:1 dr; Scheme 5a). Next, we examined the synthetic potential of the resulting HPI containing α-amino allenoate. For instance, the indole N_a atom in **3aa** was methylated

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through reductive methylation using formaldehyde, acetic acid, and sodium cyanoborohydride. Product **6** was obtained in 95% yield with 92% ee (Scheme 5b). Further, a bromo-substituted derivative **3af** was subjected to Suzuki−Miyaura coupling, to produce compound **7** in 61% isolated yield, without loss of enantiopurity (91% ee; Scheme 5c). Besides this, the allene moiety of **3aa** was subjected to hydrogenation with 10% Pd/C catalyst; the product **8** was obtained in 92% yield, 92% ee, and 8:1 dr (Scheme 5d).

CONCLUSIONS

In conclusion, we have developed a chiral Brønsted acid-catalyzed enantioselective cascade dearomatization−cyclization reaction of tryptamines with β,γ-alkynyl-α-imino esters. The γ-addition to β,γ-alkynyl-α-imino esters led to the synthesis of tetrasubstituted α-amino allenoates bearing a hexahydropyrrolo[2,3-*b*]indole unit with both axial and central chirality. The densely substituted highly enantioenriched allenes were obtained in high yields and excellent stereoselectivities. Considering the presence of both tetrasubstituted allene and HPI frameworks in a broad range of pharmaceutically potent molecules, we believe that the obtained products will find application in different disciplines. The scale-up reaction and interesting transformations further depict the practical usefulness of this strategy.

EXPERIMENTAL SECTION

Materials and Methods. The chiral phosphoric acid catalyst **PA1** is commercially available. The chiral phosphoric acid catalyst **PA2-PA7** used was synthesized according to the established literature procedure.[15] β,γ-Alkynyl-α-imino esters (**1a−r**), tryptamines (**2a−q**), alkynyl trifluoromethyl ketimine **4**, and N_a-protected tryptamine **2′** were prepared according to the modified literature procedures.[16,17] Tryptophol **2r** is commercially available. All reactions were carried out in oven-dried glassware with magnetic stirring. For heating reactions, an oil bath was used in combination with a hot plate magnetic stirrer. ^{1}H NMR and ^{13}C{^{1}H} NMR spectra were recorded on a 400 or 500 MHz JEOL NMR spectrometer in $CDCl_3$ using TMS or residual solvent signals as the internal standard. ^{19}F{^{1}H} NMR spectra were recorded on a 500 MHz JEOL NMR spectrometer in $CDCl_3$. Chemical shifts are reported in delta (δ) units, in parts per million (ppm). Coupling constants were reported in Hz. Splitting patterns are designated as s for singlet; d for doublet; t for triplet; q for quartet; dd for doublet of doublet; and m for multiplet. Mass spectrometric analysis was done by the ESI-TOF method. Routine monitoring of reactions was performed using precoated silica gel TLC plates from E-Merck. All the chromatographic separations were carried out by using silica gel (Merck's, 100−200 mesh for starting material, 230−400 mesh for product). Enantiomeric excess was determined by HPLC analysis on Daicel chiral columns using isopropanol and *n*-hexane as eluent at 25 °C. Optical rotations were measured on a commercially available automatic polarimeter. Melting points were recorded on digital melting point apparatus. Compound **3qa** was crystallized by slow evaporation of $CHCl_3$. The X-ray crystallography data of **3qa** were collected on a Bruker SMART APEX CCD diffractometer equipped with CRYO Industries low-temperature apparatus, and intensity data were collected using graphite-monochromated Mo Kα radiation (λ = 0.71073 Å) at 273.15 K. The CCDC number of complex **3qa** is 2350171 contained in the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/conts/retrieving.html (or from the Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB21EZ, UK; fax: (+44)1223- 336-033; or deposit@ccdc.cam.ac.uk).

General Procedure for Cascade Dearomatization−Cyclization Reaction. To an oven-dried round-bottom flask, equipped with magnetic stir bar, β,γ-alkynyl-α-imino ester **1** (0.1 mmol, 1.0 equiv), *N*-protected tryptamine **2** (0.12 mmol, 1.2 equiv), chiral phosphoric acid catalyst **PA5** (10 mol %), and 4 Å MS (100 mg), was added 1,2-dichloroethane (2 mL). The reaction mixture was stirred at −10 °C for 10 h. The reaction was monitored using TLC. After complete consumption of starting material, mixture was concentrated in a rotary evaporator and purified by column chromatography to afford pure product **3.**

Scale-up Reaction. To an oven-dried round-bottom flask, equipped with a magnetic stir bar, β,γ-alkynyl-α-imino ester **1a** (1 mmol, 1.0 equiv, 300 mg), *N*-tosyl-protected tryptamine **2a** (1.2 mmol, 1.2 equiv, 377 mg), chiral phosphoric acid catalyst **PA5** (10 mol %), and 4 Å MS (500 mg), was added 1,2-dichloroethane (10 mL). The reaction mixture was stirred at −10 °C for 12 h. The reaction was monitored using TLC. After complete consumption of starting material, the mixture was concentrated in a rotary evaporator and purified by column chromatography to afford pure product **3aa** with 92% yield (566 mg).

General Procedure for Racemic Reactions. Racemic BINOL phosphoric acid (10 mol %) was used for the synthesis of racemic compounds. However, these compounds were obtained in moderate yields. The racemic compounds were obtained as a mixture of diastereomers. Some of the diastereomeric mixtures were inseparable in silica gel column chromatography.

Experimental Data and Characterization. *Ethyl (R)-2-((tert-Butoxycarbonyl)amino)-4-phenyl-4-((3aR,8aS)-1-tosyl-2,3,8,8a-tetrahydropyrrolo[2,3-b]indol-3a(1H)-yl)buta-2,3-dienoate (**3aa**).* White solid, 56.58 mg, 92% yield. R_f = 0.54 (15% EtOAc in hexanes). MP: 166−168 °C. dr = >20:1. $[\alpha]_D^{20}$ = −46.78 ($CHCl_3$, *c* = 0.30 for 96% ee). HPLC (Chiralpak ID, *n*-hexane/2-propanol = 90/10, 1.0 mL/min, 254 nm): t_R = 65.77 min (major), 38.61 min (minor). ^{1}H NMR (500 MHz, $CDCl_3$) δ 7.60 (d, *J* = 8.2 Hz, 2H), 7.24−7.16 (m, 4H), 7.14−7.07 (m, 3H), 6.86 (d, *J* = 7.4 Hz, 2H), 6.70 (td, *J* = 7.5, 1.0 Hz, 1H), 6.58 (d, *J* = 7.8 Hz, 1H), 6.19 (s, 1H), 5.40 (d, *J* = 1.6 Hz, 1H), 4.80 (s, 1H), 4.28 (dq, *J* = 10.8, 7.1 Hz, 1H), 4.16 (dq, *J* = 10.8, 7.1 Hz, 1H), 3.49 (dt, *J* = 9.3, 6.6 Hz, 1H), 3.37−3.32 (m, 1H), 2.57 (dt, *J* = 12.8, 6.5 Hz, 1H), 2.41 (s, 3H), 2.25−2.18 (m, 1H), 1.48 (s, 9H), 1.29 (t, *J* = 7.1 Hz, 3H). ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$) δ 200.2, 164.7, 151.4, 148.6, 143.1, 135.6, 135.5, 130.8, 129.8, 129.1, 128.6, 128.2, 127.7, 127.4, 124.1, 118.9, 118.8, 109.4, 104.0, 81.9, 80.6, 62.5, 62.2, 47.1, 38.0, 28.3, 21.6, 14.2. HRMS (ESI) *m*/*z*: [M + H]$^+$ calcd for $C_{34}H_{38}N_3O_6S$ 616.2476; found: 616.2478.

*Ethyl (R)-2-((tert-Butoxycarbonyl)amino)-4-(4-methoxyphenyl)-4-((3aR,8aS)-1-tosyl-2,3,8,8a-tetrahydropyrrolo[2,3-b]indol-3a-(1H)-yl)buta-2,3-dienoate (**3ba**).* White solid, 54.40 mg, 85% yield. R_f = 0.53 (15% EtOAc in hexanes). MP: 118−120 °C. dr = >20:1. $[\alpha]_D^{20}$ = −45.33 ($CHCl_3$, *c* = 0.38 for 90% ee). HPLC (Chiralpak IB, *n*-hexane/2-propanol = 95/5, 1.0 mL/min, 254 nm): t_R = 42.84 min (major), 17.84 min (minor). ^{1}H NMR (500 MHz, $CDCl_3$) δ 7.61 (d, *J* = 8.2 Hz, 2H), 7.22 (d, *J* = 8.0 Hz, 3H), 7.09 (td, *J* = 7.7, 1.2 Hz, 1H), 6.80 (d, *J* = 8.7 Hz, 2H), 6.70 (t, *J* = 7.4 Hz, 1H), 6.65 (d, *J* = 8.8 Hz, 2H), 6.58 (d, *J* = 7.8 Hz, 1H), 6.19 (s, 1H), 5.42 (s, 1H), 4.80 (s, 1H), 4.22 (ddq, *J* = 62.6, 10.8, 7.2 Hz, 2H), 3.75 (s, 3H), 3.47 (dt, *J* = 9.3, 6.7 Hz, 1H), 3.38−3.30 (m, 1H), 2.55 (dt, *J* = 12.7, 6.5 Hz, 1H), 2.41 (s, 3H), 2.19 (dt, *J* = 12.8, 6.5 Hz, 1H), 1.47 (s, 9H), 1.30 (t, *J* = 7.1 Hz, 3H). ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$) δ 200.0, 164.7, 159.2, 151.4, 148.6, 143.1, 135.7, 130.9, 129.8, 129.1, 127.6, 127.4, 124.1, 118.8, 118.5, 113.7, 109.3, 103.7, 81.9, 80.5, 62.5, 62.2, 55.3, 47.1, 38.1, 28.3, 21.6, 14.2. HRMS (ESI) *m*/*z*: [M + H]$^+$ calcd for $C_{35}H_{40}N_3O_7S$ 646.2581; found: 646.2586.

*Ethyl (R)-2-((tert-Butoxycarbonyl)amino)-4-(p-tolyl)-4-((3aR,8aS)-1-tosyl-2,3,8,8a-tetrahydropyrrolo[2,3-b]indol-3a(1H)-yl)buta-2,3-dienoate (**3ca**).* White solid, 55.44 mg, 88% yield. R_f = 0.56 (15% EtOAc in hexanes). MP: 146−148 °C. dr = > 20:1. $[\alpha]_D^{20}$ = −43.44 ($CHCl_3$, *c* = 0.46 for 90% ee). HPLC (Chiralpak IB, *n*-hexane/2-propanol = 95/5, 1.0 mL/min, 254 nm): t_R = 32.48 min (major), 12.64 min (minor). ^{1}H NMR (500 MHz, $CDCl_3$) δ 7.61 (d, *J* = 8.2 Hz, 2H), 7.21 (d, *J* = 8.0 Hz, 3H), 7.09 (td, *J* = 8.6, 7.7, 1.1 Hz, 1H), 6.92 (d, *J* = 7.9 Hz, 2H), 6.75 (d, *J* = 7.9 Hz, 2H), 6.70 (t, *J* = 7.3 Hz, 1H), 6.57 (d, *J* = 7.8 Hz, 1H), 6.18 (s, 1H), 5.41 (d, *J* = 1.7 Hz, 1H), 4.84−4.71 (m, 1H), 4.33−4.08 (m, 2H), 3.48 (dt, *J* = 9.3, 6.6 Hz, 1H), 3.40−3.29 (m, 1H), 2.57 (dt, *J* = 12.9, 6.6 Hz, 1H), 2.41 (s,

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3H), 2.28 (s, 3H), 2.20 (dt, J = 12.7, 6.5 Hz, 1H), 1.47 (s, 9H), 1.29 (t, J = 7.1 Hz, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 200.1, 164.7, 151.4, 148.6, 143.1, 137.5, 135.7, 132.5, 130.9, 129.8, 129.1, 128.9, 128.5, 127.4, 124.1, 118.8, 109.4, 103.8, 81.9, 80.6, 62.5, 62.2, 47.1, 38.0, 28.3, 21.6, 21.3, 14.2. HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{35}H_{40}N_3O_6S$ 630.2632; found: 630.2639.

*Ethyl (R)-2-((tert-Butoxycarbonyl)amino)-4-(4-(tert-butyl)-phenyl)-4-((3aR,8aS)-1-tosyl-2,3,8,8a-tetrahydropyrrolo[2,3-b]-indol-3a(1H)-yl)buta-2,3-dienoate (**3da**).* White solid, 52.26 mg, 67% yield. R_f = 0.52 (15% EtOAc in hexanes). MP: 175−177 °C. dr = >20:1. $[\alpha]_D^{20}$ = −36.18 ($CHCl_3$, c = 0.39 for 87% ee). HPLC (Chiralpak IB, n-hexane/2-propanol = 92/8, 1.0 mL/min, 254 nm): t_R = 18.90 min (major), 8.26 min (minor). ^1H NMR (500 MHz, $CDCl_3$) δ 7.60 (d, J = 8.2 Hz, 2H), 7.24 (s, 1H), 7.20 (d, J = 8.1 Hz, 2H), 7.12 (d, J = 8.4 Hz, 2H), 7.09 (dd, J = 7.7, 1.2 Hz, 1H), 6.78 (d, J = 8.3 Hz, 2H), 6.69 (td, J = 7.4, 0.9 Hz, 1H), 6.59 (d, J = 7.8 Hz, 1H), 6.19 (s, 1H), 5.42 (d, J = 2.2 Hz, 1H), 4.80 (d, J = 2.2 Hz, 1H), 4.37−4.06 (m, 2H), 3.51 (dt, J = 9.2, 6.8 Hz, 1H), 3.42−3.32 (m, 1H), 2.58 (dt, J = 12.4, 6.3 Hz, 1H), 2.42 (s, 3H), 2.20 (dt, J = 13.2, 6.8 Hz, 1H), 1.48 (s, 9H), 1.30 (t, J = 7.1 Hz, 3H), 1.27 (s, 9H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 200.4, 164.7, 151.4, 150.6, 148.4, 142.9, 135.6, 132.4, 131.3, 129.8, 129.0, 128.2, 127.4, 125.2, 124.1, 118.8, 118.7, 109.3, 104.1, 81.9, 80.5, 62.5, 62.2, 47.1, 38.2, 34.6, 31.3, 28.3, 21.7, 14.2. HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{38}H_{46}N_3O_6S$ 672.3102; found: 672.3116.

*Ethyl (R)-4-([1,1′-Biphenyl]-4-yl)-2-((tert-butoxycarbonyl)-amino)-4-((3aR,8aS)-1-tosyl-2,3,8,8a-tetrahydropyrrolo[2,3-b]-indol-3a(1H)-yl)buta-2,3-dienoate (**3ea**).* White solid, 60.81 mg, 88% yield. R_f = 0.58 (15% EtOAc in hexanes). MP: 146−148 °C. dr = >20:1. $[\alpha]_D^{20}$ = −34.33 ($CHCl_3$, c = 0.60 for 90% ee). HPLC (Chiralpak IB, n-hexane/2-propanol = 90/10, 1.0 mL/min, 254 nm): t_R = 20.14 min (major), 10.20 min (minor). ^1H NMR (500 MHz, $CDCl_3$) δ 7.62 (d, J = 8.1 Hz, 2H), 7.53 (d, J = 7.4 Hz, 2H), 7.42 (t, J = 7.6 Hz, 2H), 7.35 (t, J = 7.1 Hz, 3H), 7.27 (d, J = 8.4 Hz, 1H), 7.20 (d, J = 8.0 Hz, 2H), 7.12 (t, J = 7.5 Hz, 1H), 6.95 (d, J = 8.0 Hz, 2H), 6.72 (t, J = 7.4 Hz, 1H), 6.60 (d, J = 7.8 Hz, 1H), 6.22 (s, 1H), 5.47 (s, 1H), 4.83 (s, 1H), 4.35−4.11 (m, 2H), 3.55−3.47 (m, 1H), 3.37 (q, J = 6.3 Hz, 1H), 2.60 (dt, J = 12.7, 6.5 Hz, 1H), 2.38 (s, 3H), 2.24 (dt, J = 12.7, 6.5 Hz, 1H), 1.49 (s, 9H), 1.31 (t, J = 7.1 Hz, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 200.5, 164.6, 151.4, 148.6, 143.1, 140.6, 140.5, 135.7, 134.4, 130.9, 129.8, 129.2, 129.0, 128.9, 127.5, 127.4, 127.0, 126.9, 124.1, 118.9, 118.7, 109.4, 104.2, 81.9, 80.6, 62.6, 62.2, 47.1, 38.1, 28.3, 21.6, 14.2. HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{40}H_{42}N_3O_6S$ 692.2789; found: 692.2783.

*Ethyl (R)-2-((tert-Butoxycarbonyl)amino)-4-(4-ethylphenyl)-4-((3aR,8aS)-1-tosyl-2,3,8,8a-tetrahydropyrrolo[2,3-b]indol-3a(1H)-yl)buta-2,3-dienoate (**3fa**).* White solid, 54.40 mg, 85% yield. R_f = 0.56 (15% EtOAc in hexanes). MP: 155−157 °C. dr = >20:1. $[\alpha]_D^{20}$ = −44.00 ($CHCl_3$, c = 0.35 for 92% ee). HPLC (Chiralpak IB, n-hexane/2-propanol = 95/5, 1.0 mL/min, 254 nm): t_R = 31.64 min (major), 11.90 min (minor). ^1H NMR (500 MHz, $CDCl_3$) δ 7.60 (d, J = 8.2 Hz, 2H), 7.22 (t, J = 9.4 Hz, 3H), 7.09 (td, J = 7.7, 1.2 Hz, 1H), 6.95 (d, J = 8.1 Hz, 2H), 6.78 (d, J = 8.0 Hz, 2H), 6.73−6.65 (m, 1H), 6.58 (d, J = 7.8 Hz, 1H), 6.18 (s, 1H), 5.42 (s, 1H), 4.79 (s, 1H), 4.21 (ddq, J = 65.8, 10.8, 7.1 Hz, 2H), 3.49 (dt, J = 9.3, 6.7 Hz, 1H), 3.41−3.28 (m, 1H), 2.58 (t, J = 7.4 Hz, 2H), 2.57−2.54 (m, 1H), 2.41 (s, 3H), 2.20 (dt, J = 12.8, 6.6 Hz, 1H), 1.47 (s, 9H), 1.29 (t, J = 7.1 Hz, 3H), 1.20 (t, J = 7.6 Hz, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 200.2, 164.7, 151.4, 148.5, 143.8, 143.0, 135.7, 132.7, 131.0, 129.8, 129.0, 128.5, 127.8, 127.4, 124.1, 118.8, 118.7, 109.3, 103.9, 81.9, 80.5, 62.5, 62.2, 47.1, 38.1, 28.6, 28.3, 21.6, 15.4, 14.2. HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{36}H_{42}N_3O_6S$ 644.2789; found: 644.2783.

*Ethyl (R)-2-((tert-Butoxycarbonyl)amino)-4-(4-pentylphenyl)-4-((3aR,8aS)-1-tosyl-2,3,8,8a-tetrahydropyrrolo[2,3-b]indol-3a(1H)-yl)buta-2,3-dienoate (**3ga**).* White solid, 50.69 mg, 74% yield. R_f = 0.56 (15% EtOAc in hexanes). MP: 158−160 °C. dr = >20:1. $[\alpha]_D^{20}$ = −41.95 ($CHCl_3$, c = 0.41 for 93% ee). HPLC (Chiralpak IB, n-hexane/2-propanol = 90/10, 1.0 mL/min, 254 nm): t_R = 13.12 min (major), 6.93 min (minor). ^1H NMR (500 MHz, $CDCl_3$) δ 7.60 (d, J = 7.9 Hz, 2H), 7.22 (t, J = 9.2 Hz, 3H), 7.10 (t, J = 8.2 Hz, 1H), 6.92 (d, J = 7.9 Hz, 2H), 6.77 (d, J = 7.3 Hz, 2H), 6.70 (t, J = 7.4 Hz, 1H), 6.58 (d, J = 7.9 Hz, 1H), 6.19 (s, 1H), 5.42 (s, 1H), 4.80 (s, 1H), 4.22 (dddd, J = 66.7, 14.3, 10.8, 7.2 Hz, 2H), 3.49 (dt, J = 9.0, 6.7 Hz, 1H), 3.44−3.28 (m, 1H), 2.58 (dd, J = 12.5, 6.2 Hz, 1H), 2.55−2.49 (m, 2H), 2.42 (s, 3H), 2.20 (dt, J = 13.0, 6.5 Hz, 1H), 1.58−1.54 (m, 2H), 1.48 (s, 9H), 1.34−1.27 (m, 7H), 0.89 (t, J = 6.9 Hz, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 200.2, 164.7, 151.4, 148.5, 143.0, 142.5, 135.7, 132.6, 131.1, 129.8, 129.0, 128.4, 128.3, 127.4, 124.1, 118.8, 118.7, 109.3, 103.9, 81.9, 80.5, 62.5, 62.2, 47.1, 38.1, 35.7, 31.6, 31.0, 28.3, 22.6, 21.6, 14.2, 14.0. HRMS (ESI) m/z: $[M + Na]^+$ calcd for $C_{39}H_{47}N_3NaO_6S$ 708.3078; found: 708.3070.

*Ethyl (R)-2-((tert-Butoxycarbonyl)amino)-4-(4-fluorophenyl)-4-((3aR,8aS)-1-tosyl-2,3,8,8a-tetrahydropyrrolo[2,3-b]indol-3a(1H)-yl)buta-2,3-dienoate (**3ha**).* White solid, 57.33 mg, 91% yield. R_f = 0.44 (15% EtOAc in hexanes). MP: 144−146 °C. dr = >20:1. $[\alpha]_D^{20}$ = −50.46 ($CHCl_3$, c = 0.49 for 91% ee). HPLC (Chiralpak IB, n-hexane/2-propanol = 90/10, 1.0 mL/min, 254 nm): t_R = 14.36 min (major), 7.93 min (minor). ^1H NMR (500 MHz, $CDCl_3$) δ 7.61 (d, J = 8.2 Hz, 2H), 7.22 (d, J = 8.0 Hz, 2H), 7.20 (d, J = 7.3 Hz, 1H), 7.10 (td, J = 7.7, 1.2 Hz, 1H), 6.88−6.76 (m, 4H), 6.73−6.68 (m, 1H), 6.57 (d, J = 7.8 Hz, 1H), 6.19 (s, 1H), 5.33 (d, J = 1.8 Hz, 1H), 4.82−4.70 (m, 1H), 4.23 (ddq, J = 54.6, 10.8, 7.2 Hz, 2H), 3.47 (dt, J = 9.4, 6.6 Hz, 1H), 3.37−3.29 (m, 1H), 2.52 (dd, J = 12.7, 6.4 Hz, 1H), 2.41 (s, 3H), 2.20 (dt, J = 12.7, 6.4 Hz, 1H), 1.48 (s, 9H), 1.30 (t, J = 7.1 Hz, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 200.1, 164.5, 162.3 (d, J = 247.4 Hz), 151.4, 148.6, 143.2, 135.7, 131.4, 130.5, 130.3 (d, J = 8.0 Hz), 129.8, 129.2, 127.4, 124.0, 118.9, 117.9, 115.2 (d, J = 21.3 Hz), 109.4, 104.1, 81.7, 80.7, 62.6, 62.2, 47.1, 37.9, 28.2, 21.6, 14.2. $^{19}F\{^1H\}$ NMR (471 MHz, $CDCl_3$) δ −113.95. HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{34}H_{37}FN_3O_6S$ 634.2382; found: 634.2383.

*Ethyl (R)-2-((tert-Butoxycarbonyl)amino)-4-(4-chlorophenyl)-4-((3aR,8aS)-1-tosyl-2,3,8,8a-tetrahydropyrrolo[2,3-b]indol-3a(1H)-yl)buta-2,3-dienoate (**3ia**).* White solid, 58.50 mg, 90% yield. R_f = 0.53 (15% EtOAc in hexanes). MP: 120−122 °C. dr = >20:1. $[\alpha]_D^{20}$ = −49.27 ($CHCl_3$, c = 0.44 for 92% ee). HPLC (Chiralpak IB, n-hexane/2-propanol = 90/10, 1.0 mL/min, 254 nm): t_R = 14.03 min (major), 7.76 min (minor). ^1H NMR (500 MHz, $CDCl_3$) δ 7.61 (d, J = 8.1 Hz, 2H), 7.21 (dd, J = 14.4, 7.7 Hz, 3H), 7.09 (dd, J = 18.5, 7.9 Hz, 3H), 6.79 (d, J = 8.3 Hz, 2H), 6.71 (t, J = 7.4 Hz, 1H), 6.58 (d, J = 7.8 Hz, 1H), 6.20 (s, 1H), 5.33 (s, 1H), 4.79 (s, 1H), 4.34−4.10 (m, 2H), 3.47 (dt, J = 9.0, 6.6 Hz, 1H), 3.35 (q, J = 6.3 Hz, 1H), 2.52 (dt, J = 12.8, 6.5 Hz, 1H), 2.42 (s, 3H), 2.20 (dt, J = 12.6, 6.4 Hz, 1H), 1.48 (s, 9H), 1.29 (t, J = 7.1 Hz, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 200.4, 164.4, 151.4, 148.6, 143.2, 135.6, 133.9, 133.7, 130.4, 129.9, 129.8, 129.3, 128.4, 127.4, 123.9, 118.9, 117.9, 109.4, 104.3, 81.7, 80.8, 62.6, 62.1, 47.1, 37.9, 28.2, 21.6, 14.2. HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{34}H_{37}ClN_3O_6S$ 650.2086; found: 650.2088.

*Ethyl (R)-4-(4-Bromophenyl)-2-((tert-butoxycarbonyl)amino)-4-((3aR,8aS)-1-tosyl-2,3,8,8a-tetrahydropyrrolo[2,3-b]indol-3a(1H)-yl)buta-2,3-dienoate (**3ja**).* White solid, 57.52 mg, 83% yield. R_f = 0.50 (15% EtOAc in hexanes). MP: 140−142 °C. dr = >20:1. $[\alpha]_D^{20}$ = −43.08 ($CHCl_3$, c = 0.32 for 90% ee). HPLC (Chiralpak IB, n-hexane/2-propanol = 90/10, 1.0 mL/min, 254 nm): t_R = 14.83 min (major), 8.07 min (minor). ^1H NMR (500 MHz, $CDCl_3$) δ 7.60 (d, J = 8.2 Hz, 2H), 7.25−7.21 (m, 4H), 7.19 (d, J = 7.4 Hz, 1H), 7.10 (td, J = 7.7, 1.1 Hz, 1H), 6.76−6.67 (m, 3H), 6.58 (d, J = 7.8 Hz, 1H), 6.20 (s, 1H), 5.32 (d, J = 1.6 Hz, 1H), 4.81−4.77 (m, 1H), 4.33−4.11 (m, 2H), 3.47 (dt, J = 9.4, 6.6 Hz, 1H), 3.38−3.31 (m, 1H), 2.51 (dt, J = 12.8, 6.5 Hz, 1H), 2.42 (s, 3H), 2.20 (dt, J = 12.7, 6.4 Hz, 1H), 1.48 (s, 9H), 1.29 (t, J = 7.1 Hz, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 200.3, 164.4, 151.4, 148.6, 143.3, 135.6, 134.5, 131.4, 130.4, 130.2, 129.8, 129.3, 127.4, 124.0, 121.9, 118.9, 117.9, 109.5, 104.4, 81.7, 80.8, 62.6, 62.0, 47.2, 37.9, 28.3, 21.6, 14.2. HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{34}H_{37}BrN_3O_6S$ 694.1581; found: 694.1569.

*Ethyl (R)-2-((tert-Butoxycarbonyl)amino)-4-(o-tolyl)-4-((3aR,8aS)-1-tosyl-2,3,8,8a-tetrahydropyrrolo[2,3-b]indol-3a(1H)-yl)buta-2,3-dienoate (**3ka**).* White solid, 54.81 mg, 87% yield. R_f = 0.48 (15% EtOAc in hexanes). MP: 106−108 °C. dr = >20:1. $[\alpha]_D^{20}$ = −69.30 ($CHCl_3$, c = 0.39 for 93% ee). HPLC (Chiralpak ID, n-

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hexane/2-propanol = 90/10, 1.0 mL/min, 254 nm): t_R = 54.99 min (major), 38.19 min (minor). ^{1}H NMR (500 MHz, $CDCl_3$) δ 7.57 (d, J = 8.2 Hz, 2H), 7.20 (d, J = 8.0 Hz, 2H), 7.15 (td, J = 7.5, 1.2 Hz, 1H), 7.10−7.03 (m, 2H), 6.98 (t, J = 7.4 Hz, 1H), 6.92 (d, J = 7.3 Hz, 1H), 6.81 (d, J = 6.8 Hz, 1H), 6.67−6.61 (m, 1H), 6.56 (d, J = 7.8 Hz, 1H), 6.02 (s, 1H), 5.29 (s, 1H), 4.74 (s, 1H), 4.18−4.04 (m, 2H), 3.46 (ddd, J = 10.5, 7.6, 3.0 Hz, 1H), 3.20 (td, J = 10.0, 5.8 Hz, 1H), 2.52 (ddd, J = 12.4, 9.7, 7.7 Hz, 1H), 2.40 (s, 3H), 2.31 (ddd, J = 12.4, 5.7, 3.0 Hz, 1H), 2.03 (s, 3H), 1.46 (s, 9H), 1.20 (t, J = 7.1 Hz, 3H). ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$) δ 198.7, 164.9, 151.3, 149.2, 143.3, 137.4, 135.8, 134.4, 130.3, 129.7, 129.4, 128.9, 128.9, 127.9, 127.1, 125.3, 124.6, 118.8, 117.5, 109.7, 102.9, 81.8, 80.4, 62.8, 62.1, 47.9, 37.1, 28.2, 21.6, 19.7, 14.1. HRMS (ESI) m/z: [M + H]$^+$ calcd for $C_{35}H_{40}N_3O_6S$ 630.2632; found: 630.2634.

*Ethyl (R)-2-((tert-Butoxycarbonyl)amino)-4-(m-tolyl)-4-((3aR,8aS)-1-tosyl-2,3,8,8a-tetrahydropyrrolo[2,3-b]indol-3a(1H)-yl)buta-2,3-dienoate (**3la**).* White solid, 57.31 mg, 91% yield. R_f = 0.46 (15% EtOAc in hexanes). MP: 176−178 °C. dr = >20:1. $[\alpha]_D^{20}$ = −58.14 ($CHCl_3$, c = 0.40 for 94% ee). HPLC (Chiralpak IB, n-hexane/2-propanol = 92/8, 1.0 mL/min, 254 nm): t_R = 23.09 min (major), 9.70 min (minor). ^{1}H NMR (500 MHz, $CDCl_3$) δ 7.60 (d, J = 8.2 Hz, 2H), 7.22 (d, J = 8.1 Hz, 3H), 7.09 (td, J = 7.7, 1.2 Hz, 1H), 7.00 (d, J = 5.0 Hz, 2H), 6.75 (s, 1H), 6.74−6.68 (m, 1H), 6.66−6.61 (m, 1H), 6.57 (d, J = 7.8 Hz, 1H), 6.18 (s, 1H), 5.41 (d, J = 2.0 Hz, 1H), 4.78 (d, J = 1.7 Hz, 1H), 4.21 (ddq, J = 58.7, 10.8, 7.1 Hz, 2H), 3.46 (dt, J = 9.4, 6.5 Hz, 1H), 3.36−3.28 (m, 1H), 2.57 (dt, J = 13.2, 6.7 Hz, 1H), 2.40 (s, 3H), 2.22 (dd, J = 12.6, 6.4 Hz, 1H), 2.19 (s, 3H), 1.47 (s, 9H), 1.30 (t, J = 7.1 Hz, 3H). ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$) δ 200.0, 164.7, 151.4, 148.7, 143.1, 137.8, 135.7, 135.4, 130.8, 129.7, 129.6, 129.1, 128.5, 128.1, 127.4, 125.4, 124.2, 118.9, 118.8, 109.3, 103.8, 81.8, 80.5, 62.4, 62.2, 47.1, 37.9, 28.2, 21.6, 21.5, 14.2. HRMS (ESI) m/z: [M + H]$^+$ calcd for $C_{35}H_{40}N_3O_6S$ 630.2632; found: 630.2631.

*Ethyl (R)-2-((tert-Butoxycarbonyl)amino)-4-(3,5-dimethoxyphenyl)-4-((3aR,8aS)-1-tosyl-2,3,8,8a-tetrahydropyrrolo[2,3-b]indol-3a-(1H)-yl)buta-2,3-dienoate (**3ma**).* White solid, 55.35 mg, 82% yield. R_f = 0.40 (15% EtOAc in hexanes). MP: 172−174 °C. dr = >20:1. $[\alpha]_D^{20}$ = −26.50 ($CHCl_3$, c = 0.41 for 85% ee). HPLC (Chiralpak IB, n-hexane/2-propanol = 90/10, 1.0 mL/min, 254 nm): t_R = 25.85 min (major), 11.58 min (minor). ^{1}H NMR (500 MHz, $CDCl_3$) δ 7.60 (d, J = 8.2 Hz, 2H), 7.26 (s, 1H), 7.21 (d, J = 8.1 Hz, 2H), 7.09 (td, J = 7.7, 1.2 Hz, 1H), 6.71 (td, J = 7.4, 0.8 Hz, 1H), 6.58 (d, J = 7.8 Hz, 1H), 6.29 (t, J = 2.3 Hz, 1H), 6.21 (s, 1H), 6.08 (d, J = 2.3 Hz, 2H), 5.41 (d, J = 2.3 Hz, 1H), 4.81 (d, J = 2.1 Hz, 1H), 4.36−4.12 (m, 2H), 3.57 (s, 6H), 3.41 (dt, J = 9.1, 6.8 Hz, 1H), 3.37−3.29 (m, 1H), 2.55 (dt, J = 12.4, 6.2 Hz, 1H), 2.39 (s, 3H), 2.20 (dt, J = 13.0, 6.7 Hz, 1H), 1.46 (s, 9H), 1.32 (t, J = 7.1 Hz, 3H). ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$) δ 199.8, 164.6, 160.4, 151.4, 148.8, 143.2, 137.5, 135.5, 131.1, 129.8, 129.2, 127.4, 124.1, 118.9, 118.8, 109.2, 106.6, 104.1, 100.4, 81.9, 80.5, 62.6, 62.2, 55.1, 46.9, 38.2, 28.2, 21.6, 14.2. HRMS (ESI) m/z: [M + H]$^+$ calcd for $C_{36}H_{42}N_3O_8S$ 676.2687; found: 676.2693.

*Ethyl (R)-2-((tert-Butoxycarbonyl)amino)-4-(naphthalen-1-yl)-4-((3aR,8aS)-1-tosyl-2,3,8,8a-tetrahydropyrrolo[2,3-b]indol-3a(1H)-yl)buta-2,3-dienoate (**3na**).* White solid, 55.88 mg, 85% yield. R_f = 0.44 (15% EtOAc in hexanes). MP: 163−165 °C. dr = >20:1. $[\alpha]_D^{20}$ = −37.09 ($CHCl_3$, c = 0.45 for 93% ee). HPLC (Chiralpak IB, n-hexane/2-propanol = 90/10, 1.0 mL/min, 254 nm): t_R = 20.25 min (major), 9.69 min (minor). ^{1}H NMR (500 MHz, $CDCl_3$) δ 7.79−7.73 (m, 1H), 7.58 (dd, J = 13.7, 6.1 Hz, 2H), 7.51 (d, J = 8.0 Hz, 2H), 7.47−7.42 (m, 2H), 7.40 (s, 1H), 7.27 (d, J = 7.2 Hz, 1H), 7.13 (td, J = 7.7, 1.1 Hz, 1H), 7.04 (d, J = 7.9 Hz, 2H), 6.99 (d, J = 8.4 Hz, 1H), 6.73 (t, J = 7.2 Hz, 1H), 6.61 (d, J = 7.8 Hz, 1H), 6.22 (s, 1H), 5.55−5.31 (m, 1H), 4.85−4.78 (m, 1H), 4.22 (ddq, J = 61.0, 10.8, 7.1 Hz, 2H), 3.50 (dt, J = 9.3, 6.6 Hz, 1H), 3.42−3.33 (m, 1H), 2.61 (dt, J = 12.8, 6.5 Hz, 1H), 2.35 (s, 3H), 2.27 (dt, J = 12.7, 6.5 Hz, 1H), 1.51 (s, 9H), 1.30 (t, J = 7.1 Hz, 3H). ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$) δ 200.7, 164.6, 151.4, 148.6, 143.0, 135.5, 133.1, 132.9, 132.7, 130.9, 129.7, 129.2, 128.3, 128.1, 127.8, 127.6, 127.3, 126.3, 126.1, 124.2, 118.9, 118.9, 109.4, 104.2, 81.9, 80.6, 62.5, 62.3, 47.2, 38.0, 28.3, 21.6, 14.2. HRMS (ESI) m/z: [M + H]$^+$ calcd for $C_{38}H_{40}N_3O_6S$ 666.2632; found: 666.2632.

*Ethyl (S)-2-((tert-Butoxycarbonyl)amino)-4-(thiophen-2-yl)-4-((3aS,8aS)-1-tosyl-2,3,8,8a-tetrahydropyrrolo[2,3-b]indol-3a(1H)-yl)buta-2,3-dienoate (**3oa**).* White solid, 53.32 mg, 86% yield. R_f = 0.38 (15% EtOAc in hexanes). MP: 159−161 °C. dr = >20:1. $[\alpha]_D^{20}$ = −64.88 ($CHCl_3$, c = 0.32 for 90% ee). HPLC (Chiralpak IC, n-hexane/2-propanol = 70/30, 1.0 mL/min, 254 nm): t_R = 75.58 min (major), 16.26 min (minor). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.68 (d, J = 8.1 Hz, 2H), 7.26 (s, 1H), 7.24 (s, 1H), 7.17 (d, J = 7.3 Hz, 1H), 7.10 (t, J = 7.0 Hz, 2H), 6.81−6.76 (m, 1H), 6.69 (t, J = 7.4 Hz, 1H), 6.64 (d, J = 3.1 Hz, 1H), 6.59 (d, J = 7.8 Hz, 1H), 6.23 (s, 1H), 5.64−5.53 (m, 1H), 4.91 (s, 1H), 4.36−4.08 (m, 2H), 3.48 (dt, J = 9.1, 6.5 Hz, 1H), 3.33 (q, J = 6.7 Hz, 1H), 2.59 (dt, J = 13.5, 7.0 Hz, 1H), 2.39 (s, 3H), 2.22 (dt, J = 12.2, 5.9 Hz, 1H), 1.44 (s, 9H), 1.29 (t, J = 7.1 Hz, 3H). ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$) δ 200.9, 164.1, 151.4, 149.0, 143.3, 136.9, 135.9, 130.3, 129.8, 129.4, 127.4, 127.1, 126.8, 126.5, 124.3, 118.9, 114.4, 109.4, 104.4, 81.9, 80.9, 62.7, 62.2, 46.9, 37.6, 28.2, 21.6, 14.2. HRMS (ESI) m/z: [M + H]$^+$ calcd for $C_{32}H_{36}N_3O_6S_2$ 622.2040; found: 622.2046.

*Ethyl (R)-2-((tert-Butoxycarbonyl)amino)-4-((R)-cyclohex-2-en-1-yl)-4-((3aR,8aS)-1-tosyl-2,3,8,8a-tetrahydropyrrolo[2,3-b]indol-3a-(1H)-yl)buta-2,3-dienoate (**3pa**).* White solid, 50.22 mg, 81% yield. R_f = 0.44 (15% EtOAc in hexanes). MP: 87−89 °C. dr = >20:1. $[\alpha]_D^{20}$ = −68.41 ($CHCl_3$, c = 0.36 for 83% ee). HPLC (Chiralpak ID, n-hexane/2-propanol = 90/10, 1.0 mL/min, 254 nm): t_R = 74.15 min (major), 37.97 min (minor). ^{1}H NMR (500 MHz, $CDCl_3$) δ 7.77 (d, J = 8.3 Hz, 2H), 7.30 (d, J = 8.0 Hz, 2H), 7.10 (d, J = 7.3 Hz, 1H), 7.04 (td, J = 7.6, 1.2 Hz, 1H), 6.66 (td, J = 7.4, 0.8 Hz, 1H), 6.56 (d, J = 7.8 Hz, 1H), 6.11 (s, 1H), 5.59 (s, 1H), 5.37 (s, 1H), 4.88 (s, 1H), 4.32−4.10 (m, 2H), 3.48 (dt, J = 9.4, 6.6 Hz, 1H), 3.36−3.24 (m, 1H), 2.51 (dt, J = 13.3, 6.8 Hz, 1H), 2.41 (s, 3H), 2.10 (dt, J = 12.6, 6.3 Hz, 1H), 1.97−1.81 (m, 2H), 1.72 (q, J = 16.8 Hz, 2H), 1.48 (d, J = 7.4 Hz, 2H), 1.46 (s, 9H), 1.43 (d, J = 5.1 Hz, 2H), 1.28 (t, J = 7.1 Hz, 3H). ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$) δ 200.5, 164.9, 151.5, 147.9, 143.3, 136.0, 131.9, 131.8, 129.8, 128.7, 128.1, 127.5, 123.9, 120.9, 118.5, 109.2, 103.6, 81.8, 80.4, 62.3, 61.2, 47.0, 38.3, 28.4, 28.3, 25.8, 22.8, 21.8, 21.6, 14.2. HRMS (ESI) m/z: [M + H]$^+$ calcd for $C_{34}H_{42}N_3O_6S$ 620.2789; found: 620.2785.

*Methyl (R)-2-((tert-Butoxycarbonyl)amino)-4-phenyl-4-((3aR,8aS)-1-tosyl-2,3,8,8a-tetrahydropyrrolo[2,3-b]indol-3a(1H)-yl)buta-2,3-dienoate (**3qa**).* White solid, 55.31 mg, 92% yield. R_f = 0.48 (15% EtOAc in hexanes). MP: 195−197 °C. dr = >20:1. $[\alpha]_D^{20}$ = −45.07 ($CHCl_3$, c = 0.36 for 94% ee). HPLC (Chiralpak IB, n-hexane/2-propanol = 85/15, 1.0 mL/min, 254 nm): t_R = 15.44 min (major), 7.32 min (minor). ^{1}H NMR (500 MHz, $CDCl_3$) δ 7.60 (d, J = 8.2 Hz, 2H), 7.21 (d, J = 8.1 Hz, 2H), 7.19−7.14 (m, 2H), 7.11 (ddd, J = 7.6, 6.5, 4.6 Hz, 3H), 6.87 (d, J = 7.4 Hz, 2H), 6.72 (td, J = 7.4, 0.9 Hz, 1H), 6.58 (d, J = 7.8 Hz, 1H), 6.18 (s, 1H), 5.41 (d, J = 2.1 Hz, 1H), 4.80 (d, J = 1.9 Hz, 1H), 3.78 (s, 3H), 3.48 (dt, J = 9.3, 6.7 Hz, 1H), 3.35 (dt, J = 9.2, 6.3 Hz, 1H), 2.56 (dt, J = 12.7, 6.4 Hz, 1H), 2.41 (s, 3H), 2.21 (dt, J = 12.9, 6.6 Hz, 1H), 1.48 (s, 9H). ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$) δ 200.5, 165.1, 151.4, 148.6, 143.1, 135.6, 135.4, 130.9, 129.8, 129.1, 128.6, 128.2, 127.8, 127.4, 123.9, 119.0, 109.4, 103.9, 81.8, 80.7, 62.1, 53.2, 47.1, 38.1, 28.2, 21.6. HRMS (ESI) m/z: [M + H]$^+$ calcd for $C_{33}H_{36}N_3O_6S$ 602.2319; found: 602.2318.

*Ethyl (R)-2-((tert-Butoxycarbonyl)amino)-6-phenyl-4-((3aR,8aS)-1-tosyl-2,3,8,8a-tetrahydropyrrolo[2,3-b]indol-3a(1H)-yl)hexa-2,3-dienoate (**3ra**).* Yellow oil, 35.81 mg, 56% yield. R_f = 0.27 (15% EtOAc in hexanes). dr = > 20:1. $[\alpha]_D^{20}$ = −85.00 ($CHCl_3$, c = 0.15 for 95% ee). HPLC (Chiralpak ID, n-hexane/2-propanol = 70/30, 1.0 mL/min, 254 nm): t_R = 35.56 min (major), 23.05 min (minor). ^{1}H NMR (500 MHz, $CDCl_3$) δ 7.74−7.69 (m, 2H), 7.55−7.47 (m, 1H), 7.27 (s, 1H), 7.21−7.15 (m, 2H), 7.14−7.07 (m, 2H), 7.04 (dd, J = 7.4, 1.3 Hz, 1H), 7.00−6.95 (m, 2H), 6.72 (td, J = 7.4, 1.0 Hz, 1H), 6.62 (d, J = 8.0 Hz, 1H), 6.08 (s, 1H), 5.41 (s, 1H), 4.92 (d, J = 3.2 Hz, 1H), 4.14 (ddq, J = 48.1, 10.9, 7.2 Hz, 2H), 3.38 (ddd, J = 10.3, 7.7, 2.9 Hz, 1H), 3.17 (td, J = 10.0, 5.9 Hz, 1H), 2.68−2.63 (m, 1H), 2.49 (ddd, J = 13.9, 10.1, 6.1 Hz, 1H), 2.38 (s, 3H), 2.37−2.24 (m, 2H), 2.17−2.04 (m, 2H), 1.44 (s, 9H), 1.23 (t, J = 7.1 Hz, 3H).

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$^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 197.0, 165.2, 151.8, 149.3, 143.4, 141.6, 136.2, 132.4, 129.9, 129.5, 129.2, 129.1, 128.3, 127.4, 127.2, 125.9, 124.0, 119.2, 118.5, 109.9, 104.7, 82.1, 80.5, 62.2, 61.6, 47.3, 36.2, 33.8, 30.8, 28.3, 21.6, 14.2. HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{36}H_{42}N_3O_6S$ 644.2789; found: 644.2796.

*Ethyl (R)-2-((tert-Butoxycarbonyl)amino)-4-((3aR,8aS)-5-methoxy-1-tosyl-2,3,8,8a-tetrahydropyrrolo[2,3-b]indol-3a(1H)-yl)-4-phenylbuta-2,3-dienoate (**3ab**).* Orange sticky solid, 50.31 mg, 78% yield. R_f = 0.31 (15% EtOAc in hexanes). dr = > 20:1. $[\alpha]_D^{20}$ = −33.76 ($CHCl_3$, c = 0.43 for 95% ee). HPLC (Chiralpak IA, n-hexane/2-propanol = 90/10, 1.0 mL/min, 254 nm): t_R = 61.46 min (major), 14.98 min (minor). ^1H NMR (500 MHz, $CDCl_3$) δ 7.60 (d, J = 8.2 Hz, 2H), 7.23−7.17 (m, 3H), 7.13 (t, J = 7.5 Hz, 2H), 6.90 (d, J = 7.4 Hz, 2H), 6.86−6.81 (m, 1H), 6.67 (dd, J = 8.5, 2.6 Hz, 1H), 6.51 (d, J = 8.5 Hz, 1H), 6.18 (s, 1H), 5.40 (s, 1H), 4.55 (s, 1H), 4.30−4.13 (m, 2H), 3.69 (s, 3H), 3.49 (dt, J = 9.4, 6.5 Hz, 1H), 3.32 (dt, J = 9.5, 6.5 Hz, 1H), 2.53 (dt, J = 13.4, 6.8 Hz, 1H), 2.41 (s, 3H), 2.21 (dt, J = 12.5, 6.3 Hz, 1H), 1.47 (s, 9H), 1.28 (t, J = 7.1 Hz, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 200.4, 164.6, 153.9, 151.4, 143.1, 142.7, 135.8, 135.4, 132.4, 129.8, 128.6, 128.3, 127.7, 127.4, 118.7, 113.8, 111.6, 110.1, 103.9, 82.6, 80.6, 62.6, 62.5, 56.1, 47.2, 37.7, 28.3, 21.6, 14.1. HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{35}H_{40}N_3O_7S$ 646.2581; found: 646.2583.

*Ethyl (R)-2-((tert-Butoxycarbonyl)amino)-4-((3aR,8aS)-5-methyl-1-tosyl-2,3,8,8a-tetrahydropyrrolo[2,3-b]indol-3a(1H)-yl)-4-phenylbuta-2,3-dienoate (**3ac**).* White solid, 47.18 mg, 75% yield. R_f = 0.33 (15% EtOAc in hexanes). MP: 156−158 °C. dr = >20:1. $[\alpha]_D^{20}$ = −41.01 ($CHCl_3$, c = 0.40 for 95% ee). HPLC (Chiralpak IB, n-hexane/2-propanol = 85/15, 1.0 mL/min, 254 nm): t_R = 13.69 min (major), 6.65 min (minor). ^1H NMR (500 MHz, $CDCl_3$) δ 7.60 (d, J = 8.2 Hz, 2H), 7.21 (d, J = 8.0 Hz, 2H), 7.19−7.16 (m, 1H), 7.12 (t, J = 7.5 Hz, 2H), 7.00 (s, 1H), 6.89 (t, J = 7.6 Hz, 3H), 6.48 (d, J = 7.9 Hz, 1H), 6.19 (s, 1H), 5.38 (d, J = 1.8 Hz, 1H), 4.65 (d, J = 1.9 Hz, 1H), 4.32−4.15 (m, 2H), 3.48 (dt, J = 9.3, 6.6 Hz, 1H), 3.38−3.29 (m, 1H), 2.54 (dt, J = 12.9, 6.6 Hz, 1H), 2.40 (s, 3H), 2.22 (s, 3H), 2.21−2.15 (m, 1H), 1.48 (s, 9H), 1.30 (t, J = 7.1 Hz, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 200.3, 164.7, 151.4, 146.4, 143.0, 135.7, 135.5, 131.1, 129.8, 129.6, 128.6, 128.2, 127.7, 127.4, 124.5, 118.9, 109.4, 103.9, 82.1, 80.6, 62.4, 62.3, 47.1, 37.9, 28.3, 21.6, 20.9, 14.4. HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{35}H_{40}N_3O_6S$ 630.2632; found: 630.2648.

*Ethyl (R)-2-((tert-Butoxycarbonyl)amino)-4-((3aR,8aS)-5-fluoro-1-tosyl-2,3,8,8a-tetrahydropyrrolo[2,3-b]indol-3a(1H)-yl)-4-phenylbuta-2,3-dienoate (**3ad**).* White solid, 53.17 mg, 84% yield. R_f = 0.31 (15% EtOAc in hexanes). MP: 133−135 °C. dr = >20:1. $[\alpha]_D^{20}$ = −39.5 ($CHCl_3$, c = 0.40 for 95% ee). HPLC (Chiralpak IB, n-hexane/2-propanol = 85/15, 1.0 mL/min, 254 nm): t_R = 11.49 min (major), 6.84 min (minor). ^1H NMR (500 MHz, $CDCl_3$) δ 7.58 (d, J = 8.2 Hz, 2H), 7.20 (t, J = 7.4 Hz, 3H), 7.12 (t, J = 7.5 Hz, 2H), 6.99 (dd, J = 8.1, 2.1 Hz, 1H), 6.84 (d, J = 7.5 Hz, 2H), 6.80 (td, J = 9.0, 2.7 Hz, 1H), 6.50 (dd, J = 8.5, 4.2 Hz, 1H), 6.21 (s, 1H), 5.40 (s, 1H), 4.68 (s, 1H), 4.35−4.14 (m, 2H), 3.49 (dt, J = 9.2, 6.8 Hz, 1H), 3.41−3.33 (m, 1H), 2.56 (dt, J = 12.5, 6.3 Hz, 1H), 2.41 (s, 3H), 2.18 (dt, J = 13.4, 6.9 Hz, 1H), 1.48 (s, 9H), 1.31 (t, J = 7.1 Hz, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 200.2, 164.6, 157.2 (d, J = 235.7 Hz), 151.3, 144.5, 143.2, 135.4 (d, J = 7.5 Hz), 132.6 (d, J = 7.4 Hz), 129.8, 128.5, 128.3, 127.8, 127.4, 118.3, 115.4 (d, J = 23.3 Hz), 111.3 (d, J = 24.5 Hz), 109.8 (d, J = 8.0 Hz), 104.2, 82.4, 80.7, 62.8, 62.3, 47.1, 37.9, 28.2, 21.6, 14.1. $^{19}F\{^1H\}$ NMR (471 MHz, $CDCl_3$) δ −125.58. HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{34}H_{37}FN_3O_6S$ 634.2382; found: 634.2387.

*Ethyl (R)-2-((tert-Butoxycarbonyl)amino)-4-((3aR,8aS)-5-chloro-1-tosyl-2,3,8,8a-tetrahydropyrrolo[2,3-b]indol-3a(1H)-yl)-4-phenylbuta-2,3-dienoate (**3ae**).* White solid, 53.22 mg, 82% yield. R_f = 0.33 (15% EtOAc in hexanes). MP: 126−128 °C. dr = >20:1. $[\alpha]_D^{20}$ = −30.33 ($CHCl_3$, c = 0.38 for 94% ee). HPLC (Chiralpak IB, n-hexane/2-propanol = 85/15, 1.0 mL/min, 254 nm): t_R = 10.04 min (major), 6.62 min (minor). ^1H NMR (500 MHz, $CDCl_3$) δ 7.57 (d, J = 8.2 Hz, 2H), 7.23−7.17 (m, 4H), 7.12 (t, J = 7.5 Hz, 2H), 7.06 (dd, J = 8.4, 2.2 Hz, 1H), 6.83 (d, J = 7.4 Hz, 2H), 6.50 (d, J = 8.4 Hz, 1H), 6.22 (s, 1H), 5.38 (s, 1H), 4.81 (s, 1H), 4.28 (ddq, J = 41.9, 10.8, 7.1 Hz, 2H), 3.48 (dt, J = 9.3, 6.9 Hz, 1H), 3.38 (dt, J = 9.2, 6.2 Hz, 1H), 2.57 (dt, J = 12.3, 6.2 Hz, 1H), 2.41 (s, 3H), 2.22−2.13 (m, 1H), 1.48 (s, 9H), 1.33 (t, J = 7.1 Hz, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 200.2, 164.5, 151.3, 146.9, 143.2, 135.3, 132.9, 129.8, 129.0, 128.9, 128.5, 128.4, 127.8, 127.4, 124.1, 123.5, 118.2, 110.1, 104.3, 82.0, 80.6, 62.8, 62.1, 46.9, 38.1, 28.2, 21.6, 14.4. HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{34}H_{37}ClN_3O_6S$ 650.2086; found: 650.2084.

*Ethyl (R)-4-((3aR,8aS)-5-Bromo-1-tosyl-2,3,8,8a-tetrahydropyrrolo[2,3-b]indol-3a(1H)-yl)-2-((tert-butoxycarbonyl)-amino)-4-phenylbuta-2,3-dienoate (**3af**).* White solid, 42.97 mg, 62% yield. R_f = 0.32 (15% EtOAc in hexanes). MP: 116−118 °C. dr = >20:1. $[\alpha]_D^{20}$ = −31.33 ($CHCl_3$, c = 0.30 for 87% ee). HPLC (Chiralpak IB, n-hexane/2-propanol = 85/15, 1.0 mL/min, 254 nm): t_R = 10.31 min (major), 6.77 min (minor). ^1H NMR (500 MHz, $CDCl_3$) δ 7.57 (d, J = 8.1 Hz, 2H), 7.33 (s, 1H), 7.20 (t, J = 7.5 Hz, 4H), 7.13 (t, J = 7.5 Hz, 2H), 6.84 (d, J = 7.4 Hz, 2H), 6.46 (d, J = 8.3 Hz, 1H), 6.22 (s, 1H), 5.37 (s, 1H), 4.82 (s, 1H), 4.40−4.19 (m, 2H), 3.51−3.43 (m, 1H), 3.42−3.31 (m, 1H), 2.57 (dt, J = 12.3, 6.1 Hz, 1H), 2.41 (s, 3H), 2.17 (dt, J = 13.3, 6.8 Hz, 1H), 1.48 (s, 9H), 1.33 (t, J = 7.1 Hz, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 200.3, 164.5, 151.3, 147.4, 143.2, 135.3, 133.4, 131.8, 129.8, 128.5, 128.4, 127.8, 127.4, 126.9, 118.2, 110.6, 110.4, 104.3, 81.9, 80.6, 62.8, 62.1, 46.9, 38.2, 28.2, 21.6, 14.5. HRMS (ESI) m/z: $[M + H]^+$ calcd $C_{34}H_{37}BrN_3O_6S$ 694.1581; found: 694.1573.

*Ethyl (R)-2-((tert-Butoxycarbonyl)amino)-4-phenyl-4-((3aR,8aS)-1-(phenylsulfonyl)-2,3,8,8a-tetrahydropyrrolo[2,3-b]indol-3a(1H)-yl)buta-2,3-dienoate (**3ag**).* White solid, 51.08 mg, 85% yield. R_f = 0.50 (15% EtOAc in hexanes). MP: 167−169 °C. dr = >20:1. $[\alpha]_D^{20}$ = −50.86 ($CHCl_3$, c = 0.38 for 94% ee). HPLC (Chiralpak IB, n-hexane/2-propanol = 90/10, 1.0 mL/min, 254 nm): t_R = 13.98 min (major), 8.69 min (minor). ^1H NMR (500 MHz, $CDCl_3$) δ 7.72 (d, J = 7.8 Hz, 2H), 7.53 (t, J = 7.2 Hz, 1H), 7.43 (t, J = 7.5 Hz, 2H), 7.23 (d, J = 8.2 Hz, 1H), 7.17 (d, J = 7.2 Hz, 1H), 7.13−7.07 (m, 3H), 6.85 (d, J = 7.1 Hz, 2H), 6.71 (t, J = 7.3 Hz, 1H), 6.57 (d, J = 7.7 Hz, 1H), 6.18 (s, 1H), 5.41 (s, 1H), 4.77 (s, 1H), 4.30−4.14 (m, 2H), 3.53 (dt, J = 9.2, 6.5 Hz, 1H), 3.37 (dt, J = 12.5, 6.0 Hz, 1H), 2.57 (dt, J = 12.7, 6.5 Hz, 1H), 2.22 (dt, J = 12.5, 6.3 Hz, 1H), 1.47 (s, 9H), 1.30 (t, J = 7.1 Hz, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 200.2, 164.6, 151.4, 148.6, 138.7, 135.5, 132.5, 130.8, 129.2, 129.1, 128.5, 128.3, 127.7, 127.3, 124.1, 118.9, 118.8, 109.4, 104.0, 81.8, 80.6, 62.5, 62.2, 47.2, 37.9, 28.3, 14.2. HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{33}H_{36}N_3O_6S$ 602.2319; found: 602.2312.

*Ethyl (R)-2-((tert-Butoxycarbonyl)amino)-4-((3aR,8aS)-1-((4-methoxyphenyl)sulfonyl)-2,3,8,8a-tetrahydropyrrolo[2,3-b]indol-3a(1H)-yl)-4-phenylbuta-2,3-dienoate (**3ah**).* Orange sticky solid, 56.81 mg, 90% yield. R_f = 0.46 (15% EtOAc in hexanes). dr = >20:1. $[\alpha]_D^{20}$ = −33.82 ($CHCl_3$, c = 0.31 for 94% ee). HPLC (Chiralpak IB, n-hexane/2-propanol = 95/5, 1.0 mL/min, 254 nm): t_R = 54.18 min (major), 21.03 min (minor). ^1H NMR (500 MHz, $CDCl_3$) δ 7.65 (d, J = 8.8 Hz, 2H), 7.23−7.15 (m, 2H), 7.10 (q, J = 8.1 Hz, 3H), 6.92−6.83 (m, 4H), 6.76−6.66 (m, 1H), 6.58 (d, J = 7.8 Hz, 1H), 6.20 (s, 1H), 5.37 (s, 1H), 4.78 (s, 1H), 4.22 (ddq, J = 60.4, 10.8, 7.1 Hz, 2H), 3.86 (s, 3H), 3.49 (dt, J = 9.3, 6.7 Hz, 1H), 3.41−3.27 (m, 1H), 2.57 (dt, J = 12.6, 6.4 Hz, 1H), 2.21 (dt, J = 12.9, 6.6 Hz, 1H), 1.48 (s, 9H), 1.30 (t, J = 7.1 Hz, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 200.2, 164.7, 162.8, 151.4, 148.6, 135.5, 130.9, 130.3, 129.5, 129.1, 128.6, 128.3, 127.7, 124.1, 118.9, 118.8, 114.3, 109.4, 104.0, 81.8, 80.6, 62.5, 62.2, 55.6, 47.2, 38.1, 28.3, 14.2. HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{34}H_{38}N_3O_7S$ 632.2425; found: 632.2419.

*Ethyl (R)-2-((tert-Butoxycarbonyl)amino)-4-((3aR,8aS)-1-((4-(tert-butyl)phenyl)sulfonyl)-2,3,8,8a-tetrahydropyrrolo[2,3-b]indol-3a(1H)-yl)-4-phenylbuta-2,3-dienoate (**3ai**).* White solid, 57.18 mg, 87% yield. R_f = 0.55 (15% EtOAc in hexanes). MP: 161−163 °C. dr = >20:1. $[\alpha]_D^{20}$ = −52.73 ($CHCl_3$, c = 0.33 for 95% ee). HPLC (Chiralpak IB, n-hexane/2-propanol = 97/3, 1.0 mL/min, 254 nm): t_R = 45.53 min (major), 20.62 min (minor). ^1H NMR (500 MHz, $CDCl_3$) δ 7.65 (d, J = 8.5 Hz, 2H), 7.43 (d, J = 8.5 Hz, 2H), 7.22 (d, J = 7.2 Hz, 1H), 7.20−7.15 (m, 1H), 7.13−7.06 (m, 3H), 6.88 (d, J = 7.4 Hz, 2H), 6.75−6.68 (m, 1H), 6.56 (d, J = 7.8 Hz, 1H), 6.19 (s, 1H), 5.42 (s, 1H), 4.80 (s, 1H), 4.35−4.11 (m, 2H), 3.53 (dt, J = 9.3,

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6.5 Hz, 1H), 3.40−3.32 (m, 1H), 2.62 (dt, J = 13.2, 6.7 Hz, 1H), 2.23 (dt, J = 12.5, 6.3 Hz, 1H), 1.49 (s, 9H), 1.34 (s, 9H), 1.29 (t, J = 7.2 Hz, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 200.3, 164.7, 156.1, 151.4, 148.6, 135.8, 135.5, 130.8, 129.1, 128.6, 128.3, 127.7, 127.2, 126.1, 124.1, 118.9, 118.8, 109.3, 104.0, 81.6, 80.6, 62.5, 62.3, 47.1, 37.9, 35.2, 31.2, 28.3, 14.2. HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{37}H_{44}N_3O_6S$ 658.2945; found: 658.2958.

*Ethyl (R)-4-((3aR,8aS)-1-([1,1′-Biphenyl]-4-ylsulfonyl)-2,3,8,8a-tetrahydropyrrolo[2,3-b]indol-3a(1H)-yl)-2-((tert-butoxycarbonyl)-amino)-4-phenylbuta-2,3-dienoate (**3aj**).* White solid, 62.28 mg, 92% yield. R_f = 0.54 (15% EtOAc in hexanes). MP: 174−176 °C. dr = >20:1. $[\alpha]_D^{20}$ = −31.81 ($CHCl_3$, c = 0.43 for 94% ee). HPLC (Chiralpak IB, n-hexane/2-propanol = 90/10, 1.0 mL/min, 254 nm): t_R = 19.19 min (major), 9.82 min (minor). ^1H NMR (500 MHz, $CDCl_3$) δ 7.78 (d, J = 8.4 Hz, 2H), 7.66−7.56 (m, 4H), 7.53−7.46 (m, 2H), 7.45−7.39 (m, 1H), 7.27 (d, J = 9.4 Hz, 1H), 7.16−7.08 (m, 2H), 7.05 (t, J = 7.6 Hz, 2H), 6.86 (d, J = 7.4 Hz, 2H), 6.72 (td, J = 7.4, 0.8 Hz, 1H), 6.59 (d, J = 7.8 Hz, 1H), 6.21 (s, 1H), 5.44 (d, J = 2.1 Hz, 1H), 4.82 (d, J = 2.1 Hz, 1H), 4.22 (ddq, J = 63.1, 10.8, 7.1 Hz, 2H), 3.59 (dt, J = 9.3, 6.8 Hz, 1H), 3.48−3.38 (m, 1H), 2.62 (dt, J = 12.5, 6.3 Hz, 1H), 2.24 (dt, J = 13.0, 6.7 Hz, 1H), 1.45 (s, 9H), 1.30 (t, J = 7.1 Hz, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 200.2, 164.6, 151.4, 148.5, 145.3, 139.6, 137.2, 135.5, 130.9, 129.1, 128.5, 128.4, 128.3, 127.9, 127.8, 127.7, 127.3, 124.0, 118.9, 118.8, 109.4, 104.1, 81.9, 80.6, 62.6, 62.3, 47.3, 38.1, 28.3, 14.2. HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{39}H_{40}N_3O_6S$ 678.2632; found: 678.2627.

*Ethyl (R)-2-((tert-Butoxycarbonyl)amino)-4-((3aR,8aS)-1-((4-fluorophenyl)sulfonyl)-2,3,8,8a-tetrahydropyrrolo[2,3-b]indol-3a-(1H)-yl)-4-phenylbuta-2,3-dienoate (**3ak**).* White solid, 54.47 mg, 88% yield. R_f = 0.50 (15% EtOAc in hexanes). MP: 171−173 °C. dr = >20:1. $[\alpha]_D^{20}$ = −29.61 ($CHCl_3$, c = 0.32 for 93% ee). HPLC (Chiralpak IB, n-hexane/2-propanol = 90/10, 1.0 mL/min, 254 nm): t_R = 11.49 min (major), 7.02 min (minor). ^1H NMR (500 MHz, $CDCl_3$) δ 7.71 (ddt, J = 6.9, 5.1, 2.4 Hz, 2H), 7.27 (d, J = 7.3 Hz, 1H), 7.21−7.16 (m, 1H), 7.12 (dddd, J = 7.7, 5.8, 2.7, 1.3 Hz, 3H), 7.09−7.04 (m, 2H), 6.83 (d, J = 7.2 Hz, 2H), 6.73 (td, J = 7.4, 0.8 Hz, 1H), 6.60 (d, J = 7.8 Hz, 1H), 6.24 (s, 1H), 5.38 (d, J = 2.5 Hz, 1H), 4.77 (d, J = 2.4 Hz, 1H), 4.24 (ddq, J = 66.8, 10.9, 7.1 Hz, 2H), 3.55−3.48 (m, 1H), 3.44−3.37 (m, 1H), 2.62−2.55 (m, 1H), 2.25−2.18 (m, 1H), 1.48 (s, 9H), 1.31 (t, J = 7.1 Hz, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 200.2, 165.1 (d, J = 253.8 Hz), 164.6, 151.4, 148.3, 135.5, 134.6, 131.0, 130.0 (d, J = 9.4 Hz), 129.2, 128.6, 128.4, 127.8, 123.9, 118.9, 118.6, 116.4 (d, J = 22.4 Hz), 109.5, 104.2, 81.9, 80.7, 62.6, 62.2, 47.2, 38.2, 28.2, 14.2. $^{19}F\{^1H\}$ NMR (471 MHz, $CDCl_3$) δ −105.81. HRMS (ESI) m/z: $[M + Na]^+$ calcd for $C_{33}H_{34}FN_3NaO_6S$ 642.2045; found: 642.2039.

*Ethyl (R)-4-((3aR,8aS)-1-((4-Bromophenyl)sulfonyl)-2,3,8,8a-tetrahydropyrrolo[2,3-b]indol-3a(1H)-yl)-2-((tert-butoxycarbonyl)-amino)-4-phenylbuta-2,3-dienoate (**3al**).* White solid, 57.72 mg, 85% yield. R_f = 0.55 (15% EtOAc in hexanes). MP: 160−162 °C. dr = >20:1. $[\alpha]_D^{20}$ = −15.03 ($CHCl_3$, c = 0.37 for 90% ee). HPLC (Chiralpak IB, n-hexane/2-propanol = 90/10, 1.0 mL/min, 254 nm): t_R = 11.84 min (major), 6.90 min (minor). ^1H NMR (500 MHz, $CDCl_3$) δ 7.55 (d, J = 8.9 Hz, 2H), 7.52 (d, J = 8.9 Hz, 2H), 7.28 (d, J = 7.4 Hz, 1H), 7.22−7.17 (m, 1H), 7.16−7.09 (m, 3H), 6.84−6.78 (m, 2H), 6.73 (td, J = 7.4, 0.9 Hz, 1H), 6.60 (d, J = 7.8 Hz, 1H), 6.25 (s, 1H), 5.37 (d, J = 2.7 Hz, 1H), 4.77 (d, J = 2.6 Hz, 1H), 4.25 (ddq, J = 67.8, 10.9, 7.1 Hz, 2H), 3.53 (ddd, J = 9.1, 7.8, 6.5 Hz, 1H), 3.48−3.39 (m, 1H), 2.60 (ddd, J = 12.4, 6.3, 5.2 Hz, 1H), 2.22 (dt, J = 12.7, 7.1 Hz, 1H), 1.50 (s, 9H), 1.32 (t, J = 7.1 Hz, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 200.2, 164.6, 151.4, 148.2, 137.5, 135.6, 132.4, 131.0, 129.2, 128.9, 128.6, 128.4, 127.8, 127.5, 123.9, 118.9, 118.5, 109.5, 104.3, 82.1, 80.7, 62.7, 62.3, 47.3, 38.2, 28.3, 14.2. HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{33}H_{35}BrN_3O_6S$ 680.1424; found: 680.1424.

*Ethyl (R)-2-((tert-Butoxycarbonyl)amino)-4-((3aR,8aS)-1-((4-nitrophenyl)sulfonyl)-2,3,8,8a-tetrahydropyrrolo[2,3-b]indol-3a-(1H)-yl)-4-phenylbuta-2,3-dienoate (**3am**).* Yellow solid, 29.07 mg, 45% yield. R_f = 0.43 (15% EtOAc in hexanes). MP: 165−167 °C. dr = >20:1. $[\alpha]_D^{20}$ = +5.33 ($CHCl_3$, c = 0.38 for 91% ee). HPLC (Chiralpak IB, n-hexane/2-propanol = 90/10, 1.0 mL/min, 254 nm): t_R = 15.90 min (major), 10.52 min (minor). ^1H NMR (500 MHz, $CDCl_3$) δ 8.25−8.16 (m, 2H), 7.88−7.81 (m, 2H), 7.30 (d, J = 7.2 Hz, 1H), 7.21−7.16 (m, 1H), 7.13 (dd, J = 7.7, 1.2 Hz, 1H), 7.09 (t, J = 7.7 Hz, 2H), 6.80−6.76 (m, 2H), 6.74 (td, J = 7.5, 0.8 Hz, 1H), 6.62 (d, J = 7.8 Hz, 1H), 6.29 (s, 1H), 5.44 (d, J = 3.1 Hz, 1H), 4.79 (d, J = 2.9 Hz, 1H), 4.26 (ddq, J = 71.6, 11.1, 7.2 Hz, 2H), 3.63−3.54 (m, 1H), 3.54−3.45 (m, 1H), 2.62 (ddd, J = 12.6, 6.2, 4.6 Hz, 1H), 2.24 (ddd, J = 12.7, 8.1, 7.1 Hz, 1H), 1.48 (s, 9H), 1.32 (t, J = 7.1 Hz, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 200.2, 164.4, 151.5, 149.9, 147.9, 144.2, 135.4, 131.1, 129.4, 128.5, 128.4, 127.9, 124.4, 123.8, 119.3, 118.2, 109.6, 104.6, 82.3, 80.8, 62.8, 62.2, 47.4, 38.3, 28.2, 14.2. HRMS (ESI) m/z: $[M + Na]^+$ calcd for $C_{33}H_{34}N_4NaO_8S$ 669.1990; found: 669.1994.

*Ethyl (R)-2-((tert-Butoxycarbonyl)amino)-4-((3aR,8aS)-1-(naphthalen-2-ylsulfonyl)-2,3,8,8a-tetrahydropyrrolo[2,3-b]indol-3a(1H)-yl)-4-phenylbuta-2,3-dienoate (**3an**).* White solid, 59.81 mg, 92% yield. R_f = 0.46 (15% EtOAc in hexanes). MP: 145−147 °C. dr = >20:1. $[\alpha]_D^{20}$ = −19.86 ($CHCl_3$, c = 0.36 for 95% ee). HPLC (Chiralpak IB, n-hexane/2-propanol = 95/5, 1.0 mL/min, 254 nm): t_R = 47.20 min (major), 18.03 min (minor). ^1H NMR (500 MHz, $CDCl_3$) δ 8.26 (s, 1H), 7.92−7.84 (m, 3H), 7.73 (dd, J = 8.6, 1.6 Hz, 1H), 7.66−7.61 (m, 1H), 7.60−7.55 (m, 1H), 7.23 (d, J = 7.4 Hz, 1H), 7.13−7.07 (m, 1H), 7.04 (t, J = 7.4 Hz, 1H), 6.88 (t, J = 7.4 Hz, 2H), 6.75 (d, J = 7.5 Hz, 2H), 6.71 (t, J = 7.3 Hz, 1H), 6.59 (d, J = 7.8 Hz, 1H), 6.15 (s, 1H), 5.47 (s, 1H), 4.85 (s, 1H), 4.19 (ddq, J = 57.3, 10.9, 7.1 Hz, 2H), 3.60 (dt, J = 9.1, 6.7 Hz, 1H), 3.49−3.38 (m, 1H), 2.58 (dt, J = 12.6, 6.4 Hz, 1H), 2.23 (dt, J = 12.9, 6.7 Hz, 1H), 1.39 (s, 9H), 1.27 (t, J = 7.1 Hz, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 200.2, 164.6, 151.3, 148.5, 135.7, 135.4, 134.9, 132.2, 130.9, 129.6, 129.5, 129.1, 128.7, 128.6, 128.4, 128.1, 127.9, 127.6, 127.3, 124.1, 122.7, 118.9, 109.4, 81.9, 80.5, 62.5, 62.3, 47.3, 38.1, 28.2, 14.2. HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{37}H_{38}N_3O_6S$ 652.2476; found: 652.2488.

*Ethyl (R)-2-((tert-Butoxycarbonyl)amino)-4-phenyl-4-((3aR,8aS)-1-((2,4,6-triisopropylphenyl)sulfonyl)-2,3,8,8a-tetrahydropyrrolo-[2,3-b]indol-3a(1H)-yl)buta-2,3-dienoate (**3ao**).* White solid, 58.89 mg, 81% yield. R_f = 0.33 (15% EtOAc in hexanes). MP: 185−187 °C. dr = >20:1. $[\alpha]_D^{20}$ = −107.20 ($CHCl_3$, c = 0.50 for 97% ee). HPLC (Chiralpak ADH, n-hexane/2-propanol = 97/3, 1.0 mL/min, 254 nm): t_R = 41.03 min (major), 19.40 min (minor). ^1H NMR (500 MHz, $CDCl_3$) δ 7.25 (s, 1H), 7.19 (q, J = 5.1 Hz, 3H), 7.15−7.07 (m, 5H), 6.71 (t, J = 7.4 Hz, 1H), 6.57 (d, J = 7.8 Hz, 1H), 6.15 (s, 1H), 5.60 (s, 1H), 4.82 (s, 1H), 4.23 (dtt, J = 64.8, 13.5, 6.9 Hz, 4H), 3.28 (t, J = 6.4 Hz, 1H), 3.17 (td, J = 9.3, 5.7 Hz, 1H), 2.87 (p, J = 6.9 Hz, 1H), 2.72−2.63 (m, 1H), 2.24 (dt, J = 12.0, 4.0 Hz, 1H), 1.43 (s, 9H), 1.31 (t, J = 7.1 Hz, 3H), 1.23 (d, J = 6.9 Hz, 6H), 1.18 (t, J = 6.1 Hz, 12H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 200.7, 164.9, 153.1, 151.5, 149.4, 135.6, 131.6, 130.4, 129.1, 128.5, 128.3, 127.9, 124.4, 123.8, 119.9, 118.7, 108.8, 103.7, 80.6, 80.5, 62.4, 62.1, 45.9, 38.3, 34.2, 29.3, 28.3, 25.0, 24.8, 23.6, 23.6, 14.2. HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{42}H_{54}N_3O_6S$ 728.3728; found: 728.3720.

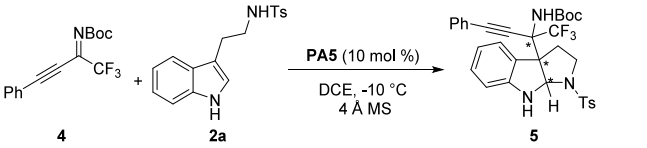
*Ethyl (R)-4-((3aR,8aS)-1-(Benzylsulfonyl)-2,3,8,8a-tetrahydropyrrolo[2,3-b]indol-3a(1H)-yl)-2-((tert-butoxycarbonyl)-amino)-4-phenylbuta-2,3-dienoate (**3ap**).* White solid, 57.20 mg, 93% yield. R_f = 0.44 (15% EtOAc in hexanes). MP: 133−135 °C. dr = >20:1. $[\alpha]_D^{20}$ = −95.07 ($CHCl_3$, c = 0.34 for 95% ee). HPLC (Chiralpak ID, n-hexane/2-propanol = 90/10, 1.0 mL/min, 254 nm): t_R = 57.40 min (major), 38.24 min (minor). ^1H NMR (500 MHz, $CDCl_3$) δ 7.37−7.32 (m, 2H), 7.32−7.26 (m, 3H), 7.23−7.13 (m, 4H), 7.11 (td, J = 7.6, 1.2 Hz, 1H), 7.06−7.01 (m, 2H), 6.74 (td, J = 7.4, 0.9 Hz, 1H), 6.55 (d, J = 7.8 Hz, 1H), 6.31 (s, 1H), 5.36 (d, J = 1.7 Hz, 1H), 4.55 (s, 1H), 4.40−4.27 (m, 2H), 4.27−4.13 (m, 2H), 3.41 (ddd, J = 9.9, 7.3, 3.8 Hz, 1H), 3.15 (q, J = 8.9 Hz, 1H), 2.71 (ddd, J = 12.4, 9.0, 7.3 Hz, 1H), 2.25−2.16 (m, 1H), 1.47 (s, 9H), 1.34 (t, J = 7.1 Hz, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 200.3, 164.8, 151.8, 149.1, 135.3, 130.9, 130.6, 129.2, 129.1, 128.8, 128.6, 128.5, 128.3, 127.9, 124.1, 119.5, 119.2, 109.6, 104.0, 81.7, 80.9, 62.6, 62.2, 58.1, 47.5, 38.2, 28.4, 14.3. HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{34}H_{38}N_3O_6S$ 616.2476; found: 616.2477.

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*tert-Butyl (3aR,8aS)-3a-((R)-3-((tert-Butoxycarbonyl)amino)-4-ethoxy-4-oxo-1-phenylbuta-1,2-dien-1-yl)-3,3a,8,8a-tetrahydropyrrolo[2,3-b]indole-1(2H)-carboxylate (**3aq**).* Colorless oil, 41.11 mg, 73% yield. R_f = 0.46 (15% EtOAc in hexanes). dr = >20:1. $[\alpha]_D^{20}$ = −161.93 ($CHCl_3$, *c* = 0.42 for 94% ee). HPLC (Chiralpak IB, *n*-hexane/2-propanol = 92/8, 1.0 mL/min, 254 nm): t_R = 13.62 min (major), 8.88 min (minor). ^{1}H NMR (500 MHz, DMSO-d_6) δ 7.62 (s, 1H), 7.24−7.14 (m, 3H), 7.14−7.02 (m, 3H), 6.98 (td, *J* = 7.6, 1.5 Hz, 1H), 6.61−6.49 (m, 2H), 6.17 (s, 1H), 5.49 (s, 1H), 4.32−4.09 (m, 2H), 3.55 (dtd, *J* = 15.4, 10.8, 9.1, 5.4 Hz, 1H), 3.04 (d, *J* = 8.8 Hz, 1H), 2.56 (dt, *J* = 12.4, 7.7 Hz, 1H), 2.09 (ddd, *J* = 11.8, 6.7, 4.5 Hz, 1H), 1.39 (d, *J* = 3.8 Hz, 18H), 1.24 (t, *J* = 7.0 Hz, 3H). ^{13}C{^{1}H} NMR (125 MHz, DMSO-d_6) δ 203.9, 164.3, 153.3, 150.3, 135.4, 131.1, 129.1, 128.7, 128.6, 128.2, 124.0, 118.7, 118.1, 109.2, 104.8, 80.4, 79.9, 79.7, 62.0, 60.6, 45.0, 38.4, 28.7, 28.5, 14.5. HRMS (ESI) *m*/*z*: [M + H]$^+$ calcd for $C_{32}H_{40}N_3O_6$ 562.2912; found: 562.2911.

*Ethyl (R)-2-((tert-Butoxycarbonyl)amino)-4-phenyl-4-((3aR,8aS)-2,3,8,8a-tetrahydro-3aH-furo[2,3-b]indol-3a-yl)buta-2,3-dienoate (**3ar**).* Yellow sticky solid, 30.49 mg, 66% yield. R_f = 0.32 (15% EtOAc in hexanes). dr = >20:1. $[\alpha]_D^{20}$ = −109.28 ($CHCl_3$, *c* = 0.38 for 87% ee). HPLC (Chiralpak IC, *n*-hexane/2-propanol = 90/20, 1.0 mL/min, 254 nm): t_R = 17.02 min (major), 6.92 min (minor). ^{1}H NMR (500 MHz, $CDCl_3$) δ 7.29 (dd, *J* = 7.5, 1.1 Hz, 1H), 7.21−7.11 (m, 6H), 7.09 (td, *J* = 7.6, 1.2 Hz, 1H), 6.73 (td, *J* = 7.4, 0.9 Hz, 1H), 6.58 (d, *J* = 7.8 Hz, 1H), 6.24 (s, 1H), 5.66 (s, 1H), 4.35 (dq, *J* = 10.8, 7.1 Hz, 1H), 4.21 (dq, *J* = 10.8, 7.1 Hz, 1H), 4.04 (ddd, *J* = 8.6, 7.1, 1.5 Hz, 1H), 3.64 (ddd, *J* = 11.2, 8.7, 4.7 Hz, 1H), 2.72 (td, *J* = 11.5, 7.2 Hz, 1H), 2.18 (ddd, *J* = 11.8, 4.7, 1.3 Hz, 1H), 1.49 (s, 9H), 1.33 (t, *J* = 7.1 Hz, 3H). ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$) δ 201.3, 164.9, 151.9, 149.7, 136.0, 131.6, 128.8, 128.7, 128.3, 128.2, 127.8, 125.0, 120.6, 119.0, 108.4, 103.8, 96.5, 81.0, 66.9, 62.5, 62.3, 41.1, 28.3, 14.3. HRMS (ESI) *m*/*z*: [M + H]$^+$ calcd for $C_{27}H_{31}N_2O_5$ 463.2227; found: 463.2221.

Control Experiments. *Reaction with Alkynyl Trifluoromethyl Ketimine **4**.*

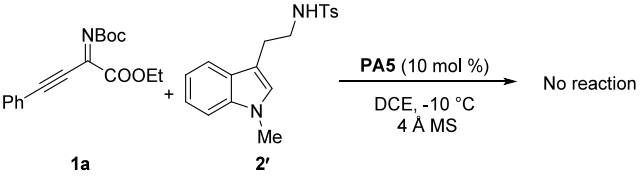


86% yield, 3% ee, >20:1 dr

To an oven-dried round-bottom flask, equipped with a magnetic stir bar, alkynyl trifluoromethyl ketimine 4 (0.1 mmol, 1.0 equiv), tryptamine 2a (0.12 mmol, 1.2 equiv), chiral phosphoric acid catalyst PA5 (10 mol %), and 4 Å MS (100 mg), was added 1,2-dichloroethane (1 mL). The reaction mixture was stirred at −10 °C for 4 h. The reaction was monitored using TLC. After complete consumption of starting material, the mixture was concentrated in a rotary evaporator and purified by column chromatography to afford pure product 5 (86% yield, 3% ee, > 20:1 dr).

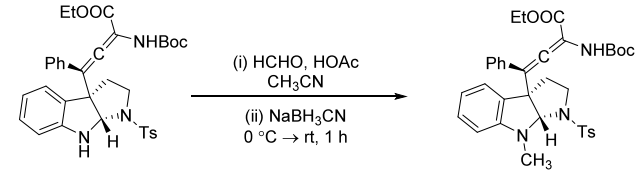
*tert-Butyl (1,1,1-Trifluoro-4-phenyl-2-(1-tosyl-2,3,8,8a-tetrahydropyrrolo[2,3-b]indol-3a(1H)-yl)but-3-yn-2-yl)carbamate (**5**).* Yellow semi solid, 52.55 mg, 86% yield. R_f = 0.53 (15% EtOAc in hexanes). dr = >20:1. HPLC (Chiralpak ID, *n*-hexane/2-propanol = 95/5, 1.0 mL/min, 254 nm): t_R = 59.34 min (major), 49.32 min (minor). ^{1}H NMR (500 MHz, $CDCl_3$) δ 7.69−7.63 (m, 2H), 7.46−7.41 (m, 2H), 7.41−7.36 (m, 1H), 7.33 (ddd, *J* = 8.3, 6.5, 1.3 Hz, 2H), 7.23 (d, *J* = 7.7 Hz, 1H), 7.18 (td, *J* = 7.7, 1.1 Hz, 1H), 7.05 (d, *J* = 8.1 Hz, 2H), 6.81 (td, *J* = 7.5, 0.8 Hz, 1H), 6.69 (d, *J* = 7.8 Hz, 1H), 5.70 (s, 1H), 5.22 (s, 1H), 5.02 (s, 1H), 3.36 (ddd, *J* = 9.9, 8.1, 1.8 Hz, 1H), 3.08 (td, *J* = 10.2, 5.8 Hz, 1H), 2.77 (ddd, *J* = 12.5, 10.4, 8.2 Hz, 1H), 2.29 (s, 3H), 2.29−2.25 (m, 1H), 1.41 (s, 9H). ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$) δ 152.9, 149.9, 143.6, 135.8, 132.1, 130.3, 129.8, 129.3, 128.4, 127.1, 126.0, 125.3, 124.4, 121.5, 119.8, 110.7, 89.3, 81.0, 79.9, 64.6, 46.4, 35.2, 28.3, 21.6. ^{19}F{^{1}H} NMR (471 MHz, $CDCl_3$) δ −72.23. HRMS (ESI) *m*/*z*: [M + H]$^+$ calcd for $C_{32}H_{33}F_3N_3O_4S$ 612.2138; found: 612.2143.

*Reaction with N_a-Protected Tryptamine **2′**.*



To an oven-dried round-bottom flask, equipped with a magnetic stir bar, *β*,*γ*-alkynyl-*α*-imino ester **1a** (0.1 mmol, 1.0 equiv), tryptamine **2′** (0.12 mmol, 1.2 equiv), chiral phosphoric acid catalyst **PA5** (10 mol %), and 4 Å MS (100 mg), was added 1,2-dichloroethane (1 mL). The reaction mixture was stirred at −10 °C for 36 h. The reaction was monitored using TLC, but the reaction did not yield any product.

Synthetic Transformations. *General Procedure and Characterization of Compound **6**.*[18]

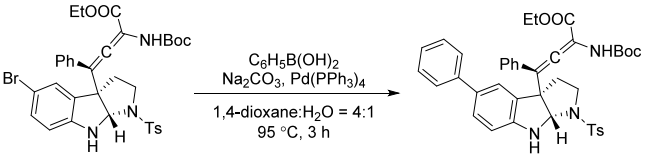


3aa: 96% ee, >20:1 dr **6**: 95% yield, 92% ee

To an oven-dried round-bottom flask, equipped with a magnetic stir bar, compound **3aa** (30.5 mg, 0.05 mmol) was taken in acetonitrile (1 mL); acetic acid (14 μL, 0.25 mmol) and formalin (HCHO aq, 37%) (19 μL, 0.25 mmol) were added. Initially, the mixture was stirred at room temperature for 15 min and then cooled to 0 °C. Then, sodium cyanoborohydride (13 mg, 0.20 mmol) was added to the reaction mixture in portions. The reaction mixture was allowed to stir at room temperature for 1 h. After complete consumption of the starting material, monitored using TLC, the reaction was quenched with $NaHCO_3$ and extracted with EtOAc. The organic phase was dried over sodium sulfate, concentrated in rotary evaporator, and purified by column chromatography to afford pure product **6** as yellow sticky solid (29.86 mg, 95% yield, 92% ee).

*Ethyl (R)-2-((tert-Butoxycarbonyl)amino)-4-((3aR,8aS)-8-methyl-1-tosyl-2,3,8,8a-tetrahydropyrrolo[2,3-b]indol-3a(1H)-yl)-4-phenylbuta-2,3-dienoate (**6**).* Yellow sticky solid, 29.86 mg, 95% yield. R_f = 0.53 (10% EtOAc in hexanes). $[\alpha]_D^{20}$ = −13.70 ($CHCl_3$, *c* = 0.36 for 92% ee). HPLC (Chiralpak IC, *n*-hexane/2-propanol = 85/15, 1.0 mL/min, 254 nm): t_R = 31.46 min (major), 20.31 min (minor). ^{1}H NMR (500 MHz, $CDCl_3$) δ 7.60−7.55 (m, 2H), 7.18 (qt, *J* = 6.3, 1.4 Hz, 4H), 7.13 (ddt, *J* = 6.6, 5.2, 1.5 Hz, 3H), 6.89−6.83 (m, 2H), 6.64 (td, *J* = 7.4, 1.0 Hz, 1H), 6.42 (d, *J* = 7.9 Hz, 1H), 6.17 (s, 1H), 5.40 (s, 1H), 4.22 (ddq, *J* = 52.3, 10.8, 7.2 Hz, 2H), 3.58 (ddd, *J* = 10.3, 6.7, 5.0 Hz, 1H), 3.22 (q, *J* = 8.2 Hz, 1H), 2.91 (s, 3H), 2.40 (s, 3H), 2.39−2.33 (m, 1H), 2.13 (dt, *J* = 12.5, 5.5 Hz, 1H), 1.50 (s, 9H), 1.29 (t, *J* = 7.1 Hz, 3H). ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$) δ 200.6, 164.8, 150.8, 143.0, 136.1, 135.5, 130.9, 129.8, 129.3, 128.6, 128.2, 127.7, 127.6, 123.8, 119.2, 117.8, 106.9, 103.9, 88.5, 80.7, 62.5, 61.5, 47.9, 38.2, 32.9, 28.4, 21.7, 14.2. HRMS (ESI) *m*/*z*: [M + H]$^+$ calcd for $C_{35}H_{40}N_3O_6S$ 630.2632; found: 630.2632.

General Procedure and Characterization of Compound 7.[11a]



3af: 87% ee, >20:1 dr **7**: 61% yield, 91% ee

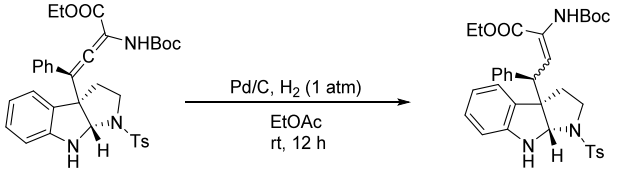
To an oven-dried round-bottom flask, equipped with a magnetic stir bar, compound **3af** (35 mg, 0.05 mmol, 1.0 equiv) was taken and treated with $Pd(PPh_3)_4$ (11.5 mg, 0.01 mmol, 20 mol %), phenylboronic acid (15 mg, 0.1 mmol, 2 equiv), and Na_2CO_3 (8 mg, 0.075 mmol, 1.5 equiv) in 1,4-dioxane/H_2O (1 mL, 4:1). The reaction mixture was stirred at 95 °C for 3 h. Upon completion of the reaction, the reaction was quenched with brine and extracted with

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DCM. The combined organic layers were dried and concentrated under reduced pressure. Then, the reaction mixture was purified using silica gel flash column chromatography to afford the product **7** as a yellowish sticky solid (21.08 mg, 61% yield, 91% ee).

*Ethyl (R)-2-((tert-Butoxycarbonyl)amino)-4-phenyl-4-((3aR,8aS)-5-phenyl-1-tosyl-2,3,8,8a-tetrahydropyrrolo[2,3-b]indol-3a(1H)-yl)-buta-2,3-dienoate (**7**).* Yellow sticky solid, 21.08 mg, 61% yield. R_f = 0.42 (15% EtOAc in hexanes). $[\alpha]_D^{20}$ = −9.31 ($CHCl_3$, c = 0.72 for 91% ee). HPLC (Chiralpak IB, *n*-hexane/2-propanol = 85/15, 1.0 mL/min, 254 nm): t_R = 15.79 min (major), 7.79 min (minor). ^{1}H NMR (500 MHz, $CDCl_3$) δ 7.62 (d, J = 8.3 Hz, 2H), 7.47−7.43 (m, 2H), 7.39 (s, 1H), 7.34 (ddd, J = 8.0, 4.7, 2.7 Hz, 3H), 7.25−7.18 (m, 4H), 7.17−7.12 (m, 2H), 6.97−6.93 (m, 2H), 6.64 (d, J = 8.1 Hz, 1H), 6.18 (s, 1H), 5.46 (s, 1H), 4.86 (s, 1H), 4.17−4.08 (m, 2H), 3.50 (dt, J = 9.5, 6.8 Hz, 1H), 3.37 (s, 1H), 2.59 (dt, J = 12.8, 6.9 Hz, 1H), 2.42 (s, 3H), 2.29 (dt, J = 12.4, 6.2 Hz, 1H), 1.46 (s, 9H), 1.07 (t, J = 7.1 Hz, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 200.5, 164.6, 151.4, 148.2, 143.2, 141.7, 135.7, 135.4, 132.6, 131.4, 129.8, 128.7, 128.6, 128.3, 128.2, 127.8, 127.4, 126.7, 126.3, 123.1, 118.8, 109.5, 104.1, 82.1, 80.6, 62.4, 62.2, 47.2, 37.7, 28.3, 21.6, 14.1. HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{40}H_{42}N_3O_6S$ 692.2789; found: 692.2798.

General Procedure and Characterization of Compound 8.



3aa: 96% ee, >20:1 dr

8: 92% yield, 92% ee 8:1 dr

To an oven-dried round-bottom flask, equipped with a magnetic stir bar, product **3aa** (31 mg, 0.05 mmol, 1.0 equiv) was taken in ethyl acetate (1 mL) and 10% Pd/C (12 mg) was added. The resulting mixture was degassed and stirred under hydrogen gas balloon pressure for 12 h at room temperature. Upon completion of the reaction, volatiles were removed under reduced pressure. The crude residue was purified using silica gel flash column chromatography to afford the product **8** as a colorless sticky solid (28.44 mg, 92% yield, 92% ee).

*Ethyl 2-((tert-Butoxycarbonyl)amino)-4-phenyl-4-((3aR,8aS)-1-tosyl-2,3,8,8a-tetrahydropyrrolo[2,3-b]indol-3a(1H)-yl)but-2-enoate (**8**).* Colorless sticky solid, 28.44 mg, 92% yield. R_f = 0.43 (15% EtOAc in hexanes). $[\alpha]_D^{20}$ = −110.90 ($CHCl_3$, c = 0.36 for 92% ee). HPLC (Chiralpak ID, *n*-hexane/2-propanol = 80/20, 1.0 mL/min, 254 nm): t_R = 38.60 min (major), 25.80 min (minor). ^{1}H NMR (500 MHz, $CDCl_3$) δ 7.64 (d, J = 8.3 Hz, 2H), 7.26 (s, 1H), 7.19−7.08 (m, 5H), 7.04 (td, J = 7.6, 1.2 Hz, 1H), 6.88 (dd, J = 7.9, 1.3 Hz, 2H), 6.76 (d, J = 7.3 Hz, 1H), 6.65 (td, J = 7.5, 0.9 Hz, 1H), 6.53 (s, 1H), 6.49 (d, J = 7.8 Hz, 1H), 5.23 (s, 1H), 4.63 (dd, J = 11.7, 2.9 Hz, 1H), 4.61 (s, 1H), 4.23 (qd, J = 7.1, 1.0 Hz, 2H), 3.39−3.31 (m, 1H), 3.06−2.97 (m, 1H), 2.41 (s, 3H), 2.18−2.09 (m, 2H), 1.47 (s, 9H), 1.29 (t, J = 7.1 Hz, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 163.6, 152.8, 149.8, 143.3, 140.1, 136.4, 129.8, 129.3, 128.9, 128.8, 128.1, 127.2, 127.0, 126.1, 125.8, 124.8, 118.9, 109.8, 82.2, 80.6, 62.5, 61.9, 49.0, 47.6, 34.3, 28.4, 21.6, 14.3. HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{34}H_{40}N_3O_6S$ 618.2632; found: 618.2638.

■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its online Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.joc.4c01021.

Crystallographic data for compound **3qa**; ^{1}H, ^{13}C{^{1}H}, and ^{19}F NMR spectra for all compounds; and HPLC chromatograms (PDF)

Accession Codes

CCDC 2350171 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

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Notes

The authors declare no competing financial interest.

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■ REFERENCES

(1) (a) Kochanowska-Karamyan, A. J.; Hamann, M. T. Marine indole alkaloids: potential new drug leads for the control of depression and anxiety. *Chem. Rev.* **2010**, *110*, 4489. (b) Zhang, Y.-C.; Jiang, F.; Shi, F. Organocatalytic Asymmetric Synthesis of Indole-Based Chiral Heterocycles: Strategies, Reactions, and Outreach. *Acc. Chem. Res.* **2020**, *53*, 425.

(2) (a) Movassaghi, M.; Schmidt, M. A.; Ashenhurst, J. A. Concise total synthesis of (+)-WIN 64821 and (−)-ditryptophenaline. *Angew. Chem., Int. Ed.* **2008**, *47*, 1485. (b) Ruiz-Sanchis, P.; Savina, S. A.; Albericio, F.; Álvarez, M. Structure, Bioactivity and Synthesis of Natural Products with Hexahydropyrrolo[2,3-*b*]indole. *Chem.—Eur. J.* **2011**, *17*, 1388.

(3) Roy, A.; Maity, A.; MK, S. S.; Giri, R.; Bisai, A. Hexahydropyrrolo-[2,3-*b*]indole alkaloids of biological relevance: proposed biosynthesis and synthetic approaches. *Arkivoc* **2020**, *2020*, 437.

(4) (a) Zheng, C.; You, S. L. Catalytic asymmetric dearomatization (CADA) reaction-enabled total synthesis of indole-based natural products. *Nat. Prod. Rep.* **2019**, *36*, 1589. (b) Sheng, F.-T.; Wang, J.-Y.; Tan, W.; Zhang, Y.-C.; Shi, F. Progresses in organocatalytic asymmetric dearomatization reactions of indole derivatives. *Org. Chem. Front* **2020**, *7*, 3967.

(5) (a) Austin, J. F.; Kim, S.-G.; Sinz, C. J.; Xiao, W.-J.; MacMillan, D. W. C. Enantioselective organocatalytic construction of pyrroloin-

dolines by a cascade addition−cyclization strategy: synthesis of (−)-flustramine B. *Proc. Natl. Acad. Sci. U. S. A.* **2004**, *101*, 5482. (b) Cai, Q.; Liu, C.; Liang, X.-W.; You, S.-L. Enantioselective construction of pyrroloindolines via chiral phosphoric acid catalyzed cascade michael addition−cyclization of tryptamines. *Org. Lett.* **2012**, *14*, 4588. (c) Duan, D.-H.; Yin, Q.; Wang, S.-G.; Gu, Q.; You, S.-L. Chiral phosphoric acid-catalyzed asymmetric cascade reaction of C(3) substituted indoles and methyl vinyl ketone. *Acta Chim. Sin.* **2014**, *72*, 1001.

(6) (a) Xie, W.; Jiang, G.; Liu, H.; Hu, J.; Pan, X.; Zhang, H.; Wan, X.; Lai, Y.; Ma, D. Highly enantioselective bromocyclization of tryptamines and its application in the synthesis of (−)-Chimonanthine. *Angew. Chem., Int. Ed.* **2013**, *52*, 12924. (b) Cai, Q.; Yin, Q.; You, S.-L. Chiral-amine-catalyzed asymmetric bromocyclization of tryptamine derivatives. *Asian J. Org. Chem.* **2014**, *3*, 408. (c) Liang, X.-W.; Liu, C.; Zhang, W.; You, S.-L. Asymmetric fluorinative dearomatization of tryptamine derivatives. *Chem. Commun.* **2017**, *53*, 5531. (d) Egami, H.; Hotta, R.; Otsubo, M.; Rouno, T.; Niwa, T.; Yamashita, K.; Hamashima, Y. Asymmetric dearomatizing fluoroamidation of indole derivatives with dianionic phase-transfer catalyst. *Org. Lett.* **2020**, *22*, 5656.

(7) (a) Zhang, Z.; Antilla, J. C. Enantioselective construction of pyrroloindolines catalyzed by chiral phosphoric acids: total synthesis of (−)-Debromoflustramine B. *Angew. Chem., Int. Ed.* **2012**, *51*, 11778. (b) Wei, Q.; Wang, Y.-Y.; Du, Y.-L.; Gong, L.-Z. Organocatalytic asymmetric selenofunctionalization of tryptamine for the synthesis of hexahydropyrrolo[2,3-*b*]indole derivatives. *Beilstein J. Org. Chem.* **2013**, *9*, 1559. (c) Nelson, H. M.; Reisberg, S. H.; Shunatona, H. P.; Patel, J. S.; Toste, F. D. Chiral anion phase transfer of aryldiazonium cations: an enantioselective synthesis of C3-diazenated pyrroloindolines. *Angew. Chem., Int. Ed.* **2014**, *53*, 5600. (d) Li, Q.; Xia, T.; Yao, L.; Deng, H.; Liao, X. Enantioselective and diastereoselective azo-coupling/iminium-cyclizations: a unified strategy for the total syntheses of (−)-psychotriasine and (+)-pestalazine B. *Chem. Sci.* **2015**, *6*, 3599.

(8) (a) Gentry, E. C.; Rono, L. J.; Hale, M. E.; Matsuura, R.; Knowles, R. R. Enantioselective Synthesis of Pyrroloindolines via Noncovalent Stabilization of Indole Radical Cations and Applications to the Synthesis of Alkaloid Natural Products. *J. Am. Chem. Soc.* **2018**, *140*, 3394. (b) Liang, K.; Tong, X.; Li, T.; Shi, B.; Wang, H.; Yan, P.; Xia, C. Enantioselective Radical Cyclization of Tryptamines by Visible Light-Excited Nitroxides. *J. Org. Chem.* **2018**, *83*, 10948. (c) Cheng, Y.; Zhao, Q.; Zhang, X.; You, S. Asymmetric dearomatization of Indole Derivatives with N-Hydroxycarbamates Enabled by Photoredox Catalysis. *Angew. Chem., Int. Ed.* **2019**, *58*, 18069.

(9) (a) Hoffmann-Röder, A.; Krause, N. Synthesis and properties of allenic natural products and pharmaceuticals. *Angew. Chem., Int. Ed.* **2004**, *43*, 1196. (b) Ma, S. Some typical advances in the synthetic applications of allenes. *Chem. Rev.* **2005**, *105*, 2829. (c) Rivera-Fuentes, P.; Diederich, F. allenes in molecular materials. *Angew. Chem., Int. Ed.* **2012**, *51*, 2818.

(10) (a) Yu, S.; Ma, S. allenes in Catalytic Asymmetric Synthesis and Natural Product Syntheses. *Angew. Chem., Int. Ed.* **2012**, *51*, 3074. (b) Alcaide, B.; Almendros, P. Progress in allene chemistry. *Chem. Soc. Rev.* **2014**, *43*, 2886. (c) Chu, W.-D.; Zhang, Y.; Wang, J. Recent advances in catalytic asymmetric synthesis of allenes. *Catal. Sci. Technol.* **2017**, *7*, 4570.

(11) (a) Wang, J.-Y.; Zhang, S.; Yu, X.-Y.; Wang, Y.-H.; Wan, H.-L.; Zhang, S.; Tan, W.; Shi, F. Organocatalytic asymmetric synthesis of bioactive hexahydropyrrolo[2,3-*b*]indole-containing tetrasubstituted allenes bearing multiple chiral elements. *Tetrahedron Chem.* **2022**, *1*, No. 100007. (b) Yue, Z.; Shen, B.; Cao, J.; Chen, X.; Fang, F.; Li, P.; Yu, P.; Li, W. Organocatalytic Enantioselective Reaction of Tertiary α-(7-indolyl)methanols with Tryptamines. *Org. Chem. Front.* **2023**, *10*, 3662.

(12) (a) Hatano, M.; Okamoto, H.; Kawakami, T.; Toh, K.; Nakatsuji, H.; Sakakura, A.; Ishihara, K. Enantioselective aza Friedel−Crafts reaction of furan with α-ketimino esters induced by a conjugated double hydrogen bond network of chiral bis(phosphoric acid) catalysts. *Chem. Sci.* **2018**, *9*, 6361. (b) Trost, B. M.; Hung, C.; Scharf, M. J. Direct Catalytic Asymmetric Vinylogous Additions of α,β- and β,γ-Butenolides to Polyfluorinated alkynyl ketimines. *Angew. Chem., Int. Ed.* **2018**, *57*, 11408. (c) Dai, J.; Wang, Z.; Deng, Y.; Zhu, L.; Peng, F.; Lan, Y.; Shao, Z. Enantiodivergence by minimal modification of an acyclic chiral secondary aminocatalyst. *Nat. Commun.* **2019**, *10*, 5182. (d) Woldegiorgis, A. G.; Han, Z.; Lin, X. X. Chiral Phosphoric Acid-Catalyzed Enantioselective Synthesis of Pyrazole-Based Unnatural α-Amino Acid Derivatives. *Adv. Synth. Catal.* **2021**, *364*, 274. (e) Ogura, K.; Takehara, T.; Suzuki, T.; Nakamura, S. Enantioselective Vinylogous Mannich Reaction of Acyclic Vinylketene Silyl Acetals with Acyclic ketimines. *Adv. Synth. Catal.* **2021**, *363*, 4544. (f) Nakamura, S.; Matsumoto, N.; Kibe, M.; Abe, K.; Takehara, T.; Suzuki, T. Enantiodivergent Reaction of ketimines with Malononitriles Using Single Cinchona Alkaloid Sulfonamide Catalysts. *Adv. Synth. Catal.* **2022**, *364*, 781. (g) Sheng, C.; Ling, Z.; Luo, Y.; Zhang, W. Cu-catalyzed asymmetric addition of alcohols to β,γ-alkynyl-α-imino esters for the construction of linear chiral *N*,*O*-ketals. *Nat. Commun.* **2022**, *13*, 400. (h) Li, M.; Chen, Y.; Yan, Y.; Liu, M.; Huang, M.; Li, W.; Cao, L.; Zhang, X. Organocatalytic asymmetric synthesis of quaternary α-isoxazole-α-alkynyl amino acid derivatives. *Org. Biomol. Chem.* **2022**, *20*, 8849. (i) Woldegiorgis, A. G.; Gu, H.; Lin, X. F. Enantioselective synthesis of indole-based unnatural β-alkynyl α-amino acid derivatives via chiral phosphoric acid catalysis. *Chirality* **2022**, *34*, 678. (j) Ji, D.-S.; Liang, H.; Yang, K.-X.; Feng, Z.-T.; Luo, Y.-C.; Xu, G.-Q.; Gu, Y.; Xu, P.-F. Solvent directed chemically divergent synthesis of β-lactams and α-amino acid derivatives with chiral isothiourea. *Chem. Sci.* **2022**, *13*, 1801.

(13) (a) Yang, J.; Wang, Z.; He, Z.; Li, G.; Hong, L.; Sun, W.; Wang, R. Organocatalytic Enantioselective Synthesis of tetrasubstituted α-Amino allenoates by Dearomative γ-Addition of 2,3-Disubstituted Indoles to β,γ-alkynyl-α-imino Esters. *Angew. Chem., Int. Ed.* **2020**, *59*, 642. (b) Woldegiorgis, A. G.; Han, Z.; Lin, X. Organocatalytic Asymmetric dearomatization Reaction for the Synthesis of Axial Chiral Allene-Derived Naphthalenones Bearing Quaternary Stereocenters. *Org. Lett.* **2021**, *23*, 6606. (c) Li, F. S.; Liang, S.; Luan, Y. P.; Chen, X. L.; Zhao, H. H.; Huang, A. Q.; Li, P. F.; Li, W. J. Organocatalytic regio-, diastereo- and enantioselective γ-additions of isoxazol-5(4*H*)-ones to β,γ-alkynyl-α-imino esters for the synthesis of axially chiral tetrasubstituted α-amino allenoates. *Org. Chem. Front.* **2021**, *8*, 1243. (d) Zhang, X.; Song, X.−X; Ni, Q.-J. Organocatalytic regio- and enantioselective C1-arylation of β,γ-alkynyl-α-imino esters with pyrrolo[2,1-*a*]isoquinolines. *Chem.Commun.* **2024**, *60*, 831.

(14) (a) Unhale, R. A.; Sadhu, M. M.; Singh, V. K. Chiral Brønsted Acid Catalyzed Enantioselective Synthesis of Spiro-Isoindolinone-Indolines via Formal [3 + 2] Cycloaddition. *Org. Lett.* **2022**, *24*, 3319. (b) Khajuria, C.; Sadhu, M. M.; Unhale, R. A.; Singh, V. K. Chiral phosphoric acid-catalyzed reaction between C-alkynyl imine precursor and thiol: Access to highly enantioenriched alkynyl isoindolinones with *N*,*S*-ketal framework. *Tetrahedron Lett.* **2022**, *112*, No. 154230. (c) Subba, P.; Sadhu, M. M.; Singh, V. K. Chiral Phosphoric Acid-Catalyzed Asymmetric Friedel−Crafts Addition of Indolizine to Cyclic *N*-Sulfonyl Imine. *J. Org. Chem.* **2023**, *88*, 14676. (d) Sahoo, S. R.; Gupta, K.; Singh, V. K. Enantioselective Dearomative [4+2] Cycloaddition Reaction of 1-Naphthols with In-Situ Generated *ortho*-Quinone Methides. *Chem.—Eur. J.* **2023**, *29*, No. e202302707. (e) Subba, P.; Sahoo, S. R.; Khajuria, C.; Singh, V. K. Enantioselective Aminative dearomatization of Indoles via Electrophilic 1,6-Addition of *p*-Quinone Diimides (*p*-QDIs). *Org. Lett.* **2024**, *26*, 4932.

(15) Featherston, A. L.; Kwon, Y.; Pompeo, M. M.; Engl, O. D.; Leahy, D. K.; Miller, S. J. Catalytic asymmetric and stereodivergent oligonucleotide synthesis. *Science* **2021**, *371*, 702.

(16) (a) Zhang, Y. L.; Duan, J. X.; Zhou, Y. Q.; Li, X. Y.; Yao, W. J.; Wang, Z. Rapid Construction of Tricyclic Furanobenzodihydropyrans by Asymmetric Tandem Reaction. *J. Org. Chem.* **2023**, *88*, 1815. (b) Sheng, C.; Ling, Z.; Ahmad, T.; Xie, F.; Zhang, W. Copper-Catalyzed Regioselective [3+3] Annulations of alkynyl ketimines with α-Cyano Ketones: the Synthesis of Polysubstituted 4*H*-Pyran

https://doi.org/10.1021/acs.joc.4c01021
J. Org. Chem. 2024, 89, 10148−10162

Derivatives with a CF_3-Containing Quaternary Center. *Chem.—Eur. J.* **2022**, *28*, No. e202200128.
(17) (a) Peng, Y.-S.; Wang, W.; Shi, J.; Wu, W.; Song, J.-R.; Pan, W.-D.; Hao, G.-F.; Ren, H. Copper-Catalyzed Oxygenative Skeletal Rearrangement of Tetrahydro-*β*-carbolines Using H_2O and O_2 as Oxygen Sources. *Angew. Chem., Int. Ed.* **2023**, *62*, No. e202313687. (b) Chen, C.; Chen, J.; Wang, H.; Xu, Z.-F.; Duan, S.; Li, C.-Y. Catalyst-Free Synthesis of Polycyclic Spiroindolines by Cascade Reaction of 3-(2-Isocyanoethyl)indoles with 1-Sulfonyl-1,2,3-triazoles. *J. Org. Chem.* **2023**, *88*, 9543.
(18) Yang, Y.; Jiang, X.; Qing, F.-L. Sequential Electrophilic Trifluoromethanesulfanylation−Cyclization of tryptamine Derivatives: Synthesis of C(3)-Trifluoromethanesulfanylated Hexahydropyrrolo[2,3-*b*]indoles. *J. Org. Chem.* **2012**, *77*, 7538.

https://doi.org/10.1021/acs.joc.4c01021
J. Org. Chem. 2024, 89, 10148−10162